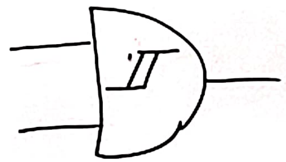
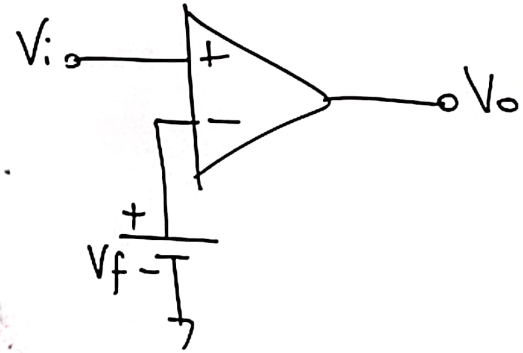
Schmitt Trigger (Tetikleyici)



Sembolü



$V_i > V_f$ ise $+V_{cc}$

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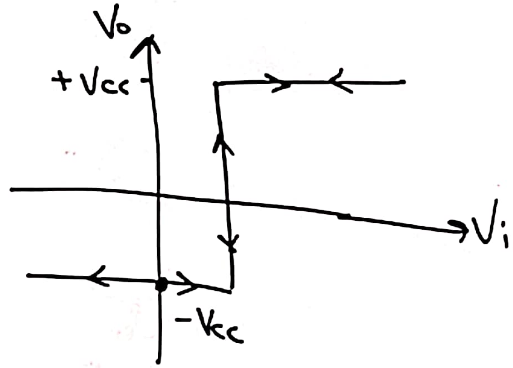

Sayısal devrelerde gürültü engellemek için kullanılır.

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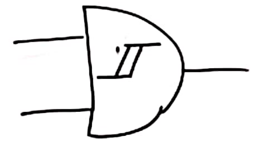

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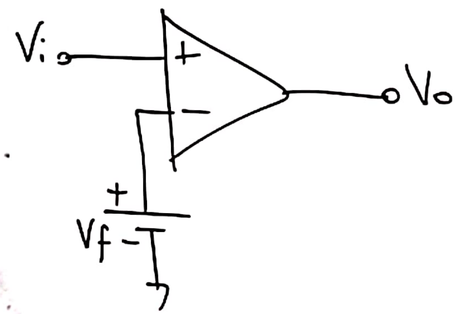

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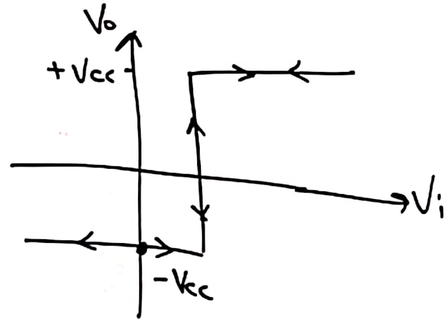

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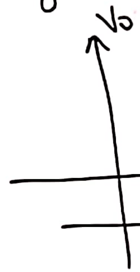
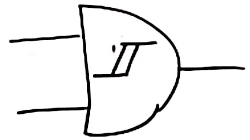
Vf gerilim

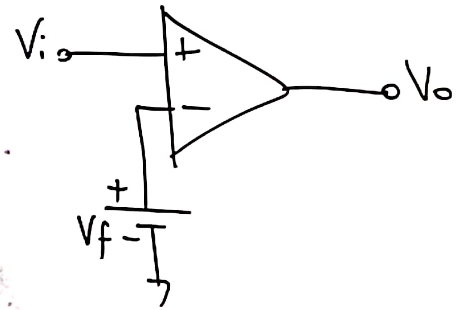


Ör

Schmitt Trigger (Tetikleyici)



Sembolü



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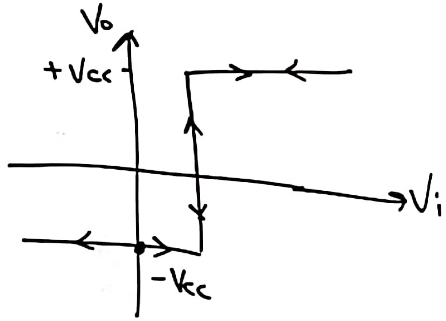

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Vf gerilim

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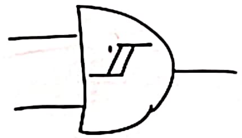
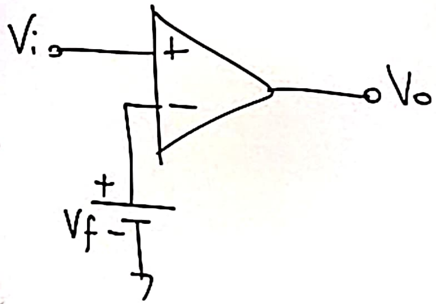


Sembolü



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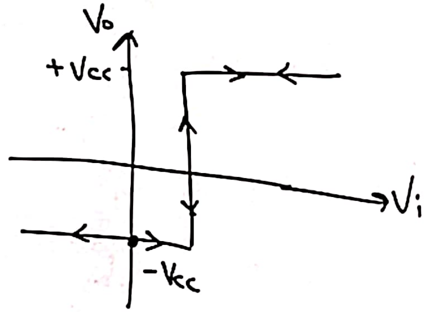

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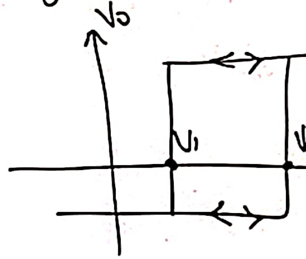
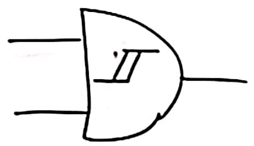
Vf gerilimi [V1-V



Örnek: Vi

∓Vc

Schmitt Trigger (Tetikleyici)



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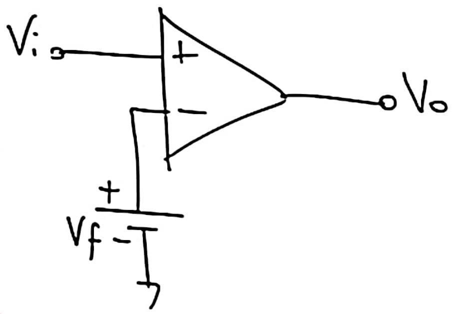

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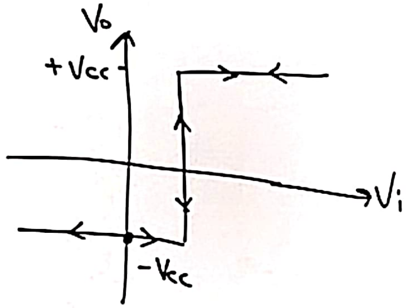

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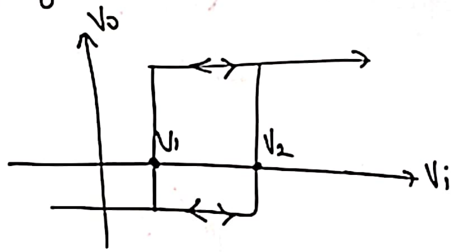
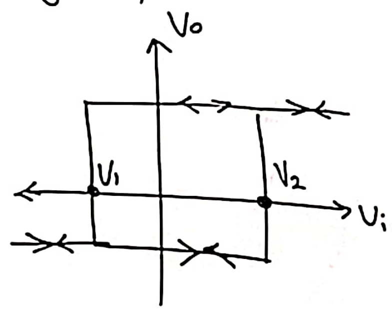
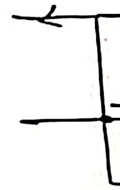
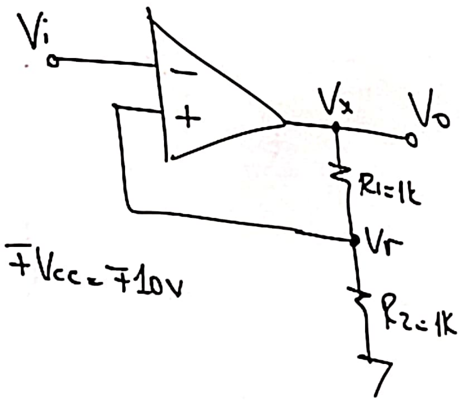
V_f gerilimi $[V_1 - V_2]$ aralığında değişirse;







Örnek:



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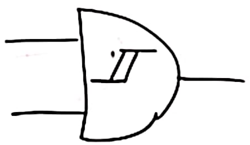
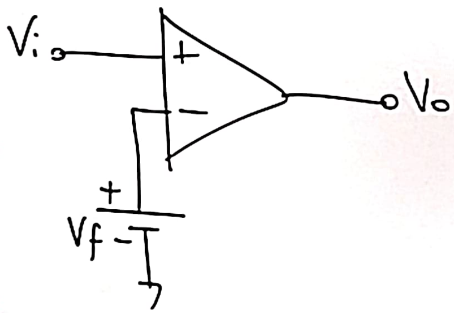


Sembolü



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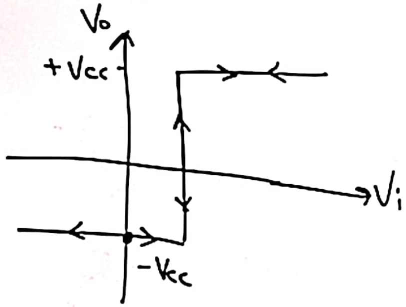

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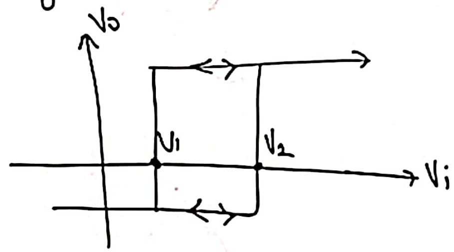
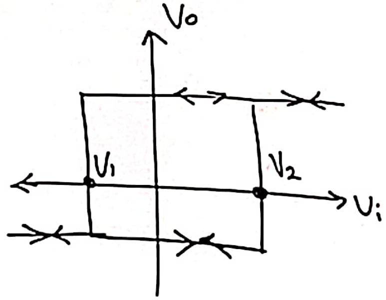
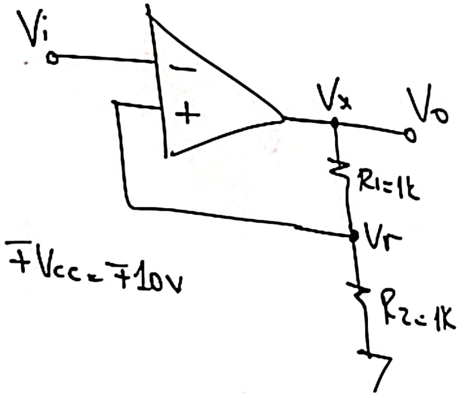




Örnek:



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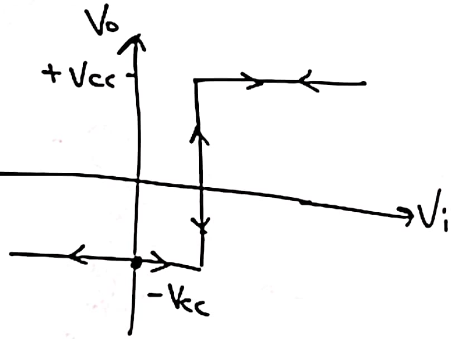

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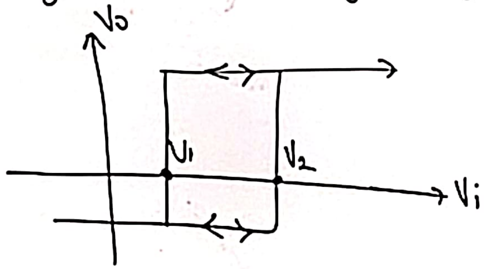

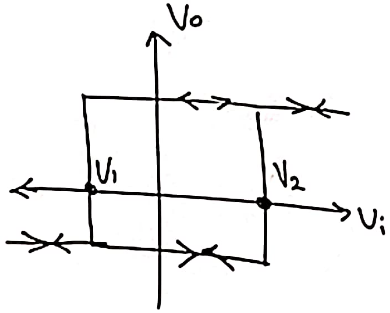

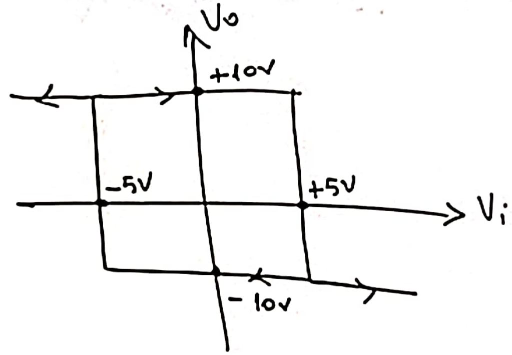

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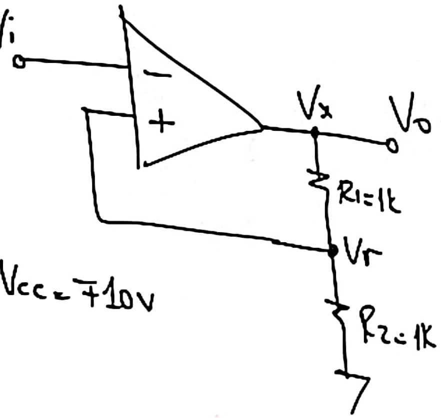

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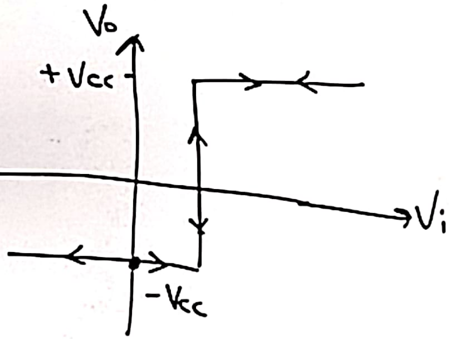

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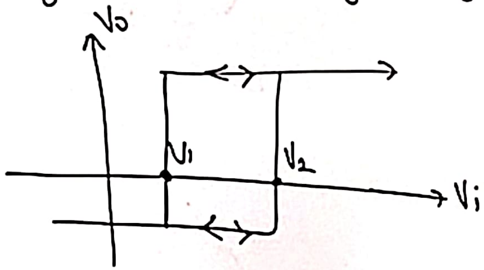

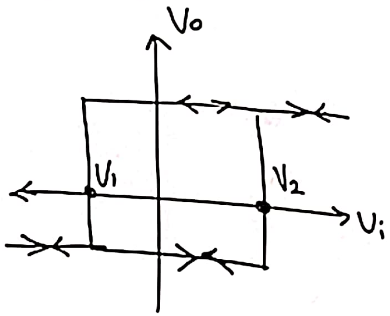

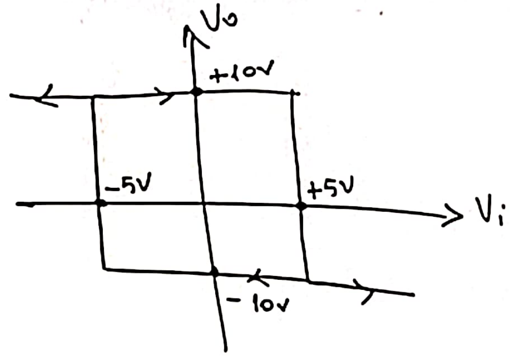

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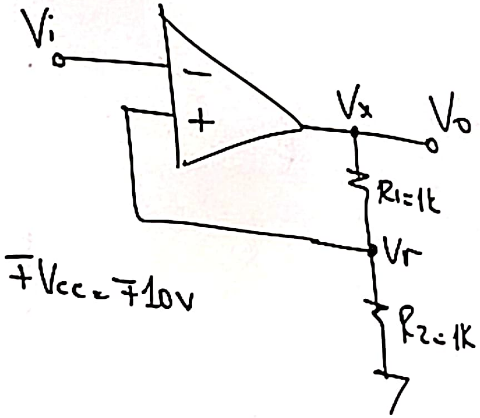

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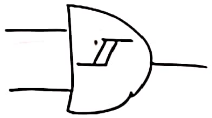
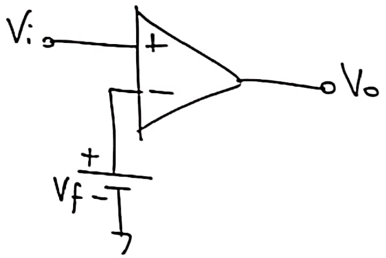


Sembolü



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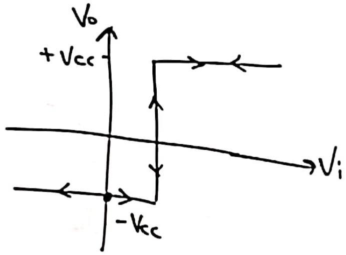

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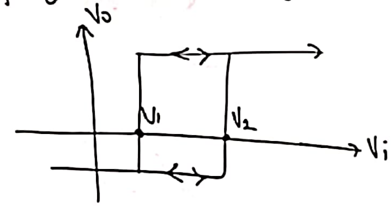
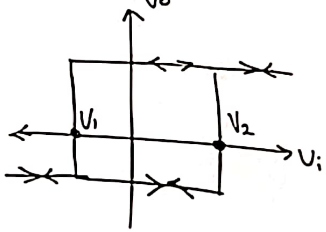
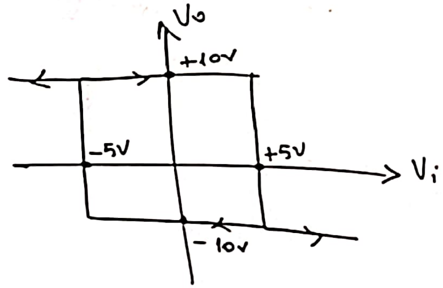
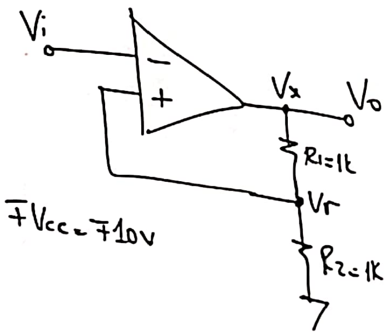






Örnek:



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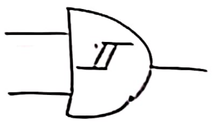
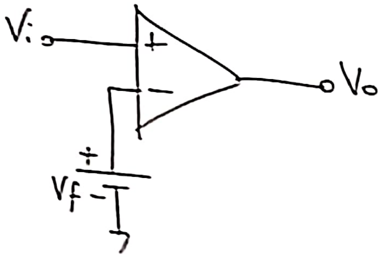


Sembolü



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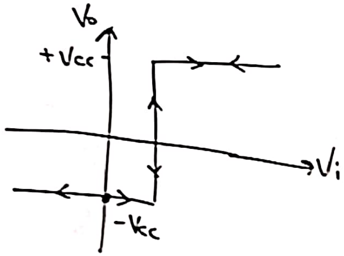

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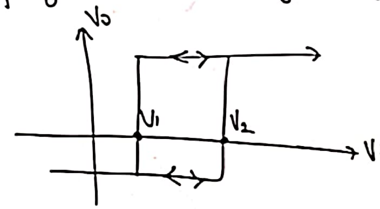
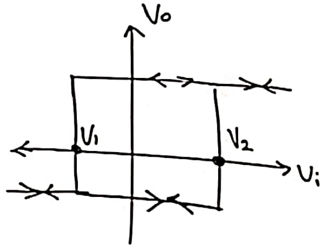
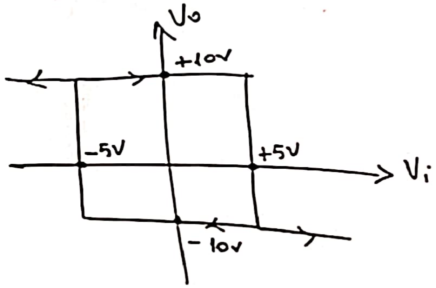
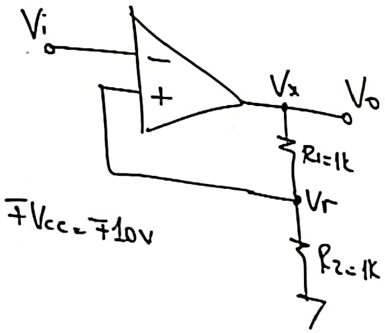






Örnek:



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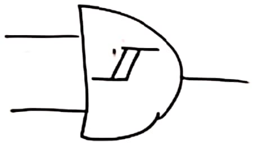
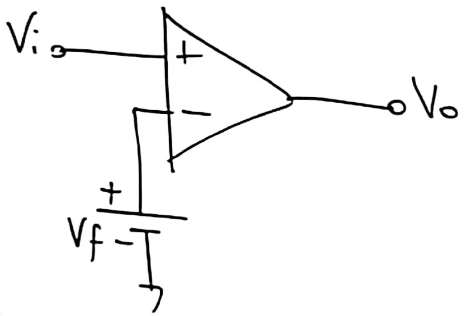


Sembolü



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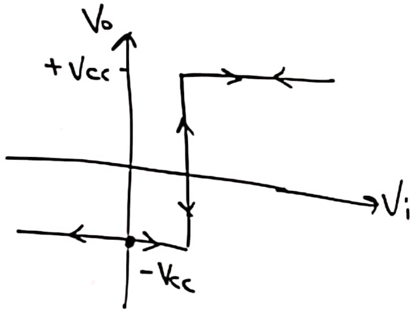

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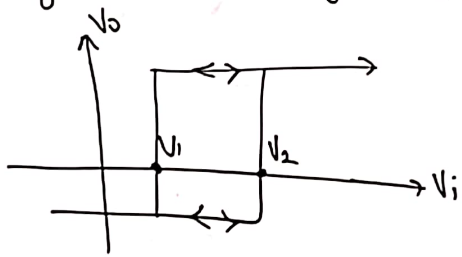

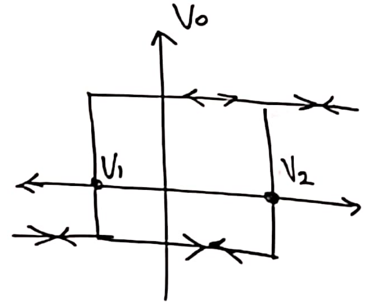

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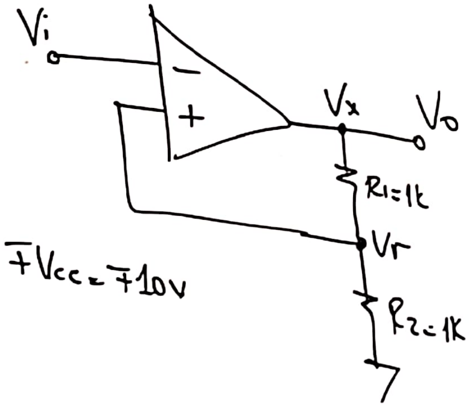

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$V_i < V_r \Rightarrow V_x = +V_{cc}$

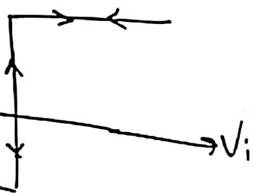
* $V_x = -V_{cc}$ ise;

$V_r = -\dfrac{V_{cc}}{2}$

...yici)

... ise $+V_{cc}$

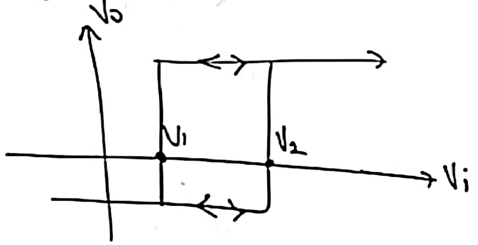
... ise $-V_{cc}$

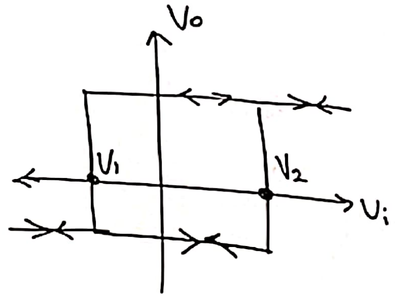


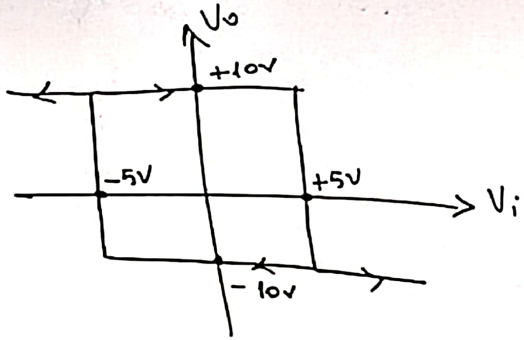
...cc

... için kullanılır.

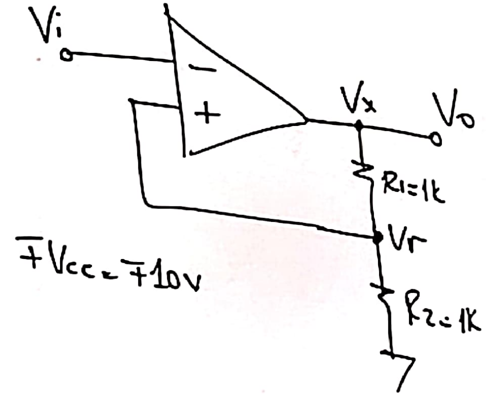
V_f gerilimi $[V_1 - V_2]$ aralığında değişirse;







Örnek:



$\mp V_{cc} = \mp 10V$

$V_r = V_x \cdot \frac{R_2}{R_1 + R_2}$

* $V_x = +V_{cc}$ ise;

$V_r = V_{cc} \cdot \frac{1k}{2k} = \frac{V_{cc}}{2}$

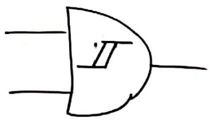
$V_i > V_r \Rightarrow V_x = -V_{cc}$

$V_i < V_r \Rightarrow V_x = +V_{cc}$

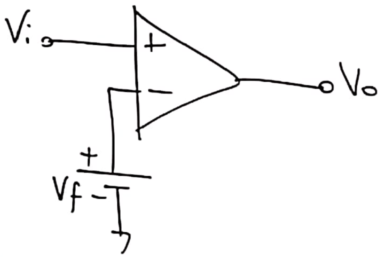
* $V_x = -V_{cc}$ ise;

$V_r = -\frac{V_{cc}}{2}$

Schmitt Trigger (Tetikleyici)

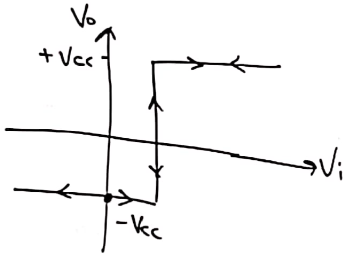


Sembolü



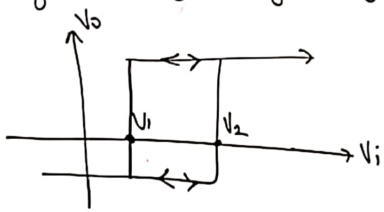
$V_i > V_f$ ise $+V_{cc}$

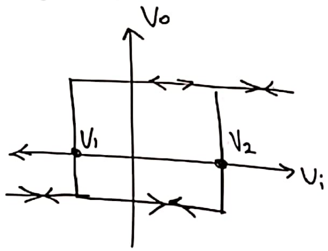
$V_i < V_f$ ise $-V_{cc}$

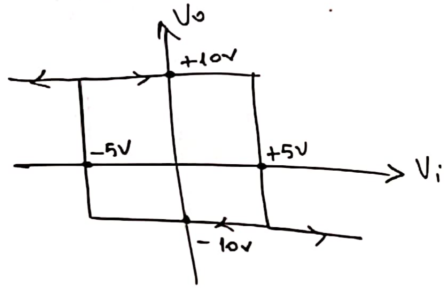


Sayısal devrelerde gürültü engellemek için kullanılır.

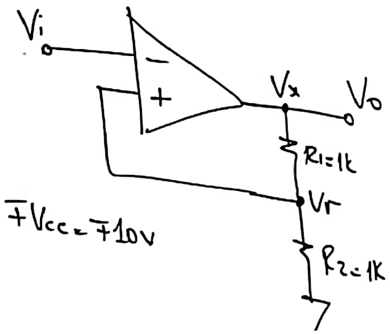
V_f gerilimi $[V_1 - V_2]$ aralığında değişirse;







Örnek:



$\mp V_{cc} = \mp 10V$

$V_r = V_x \cdot \dfrac{R_2}{R_1 + R_2}$

* $V_x = +V_{cc}$ ise;

$V_r = V_{cc} \cdot \dfrac{1k}{2k} = \dfrac{V_{cc}}{2}$

$V_i > V_r \Rightarrow V_x = -V_{cc}$

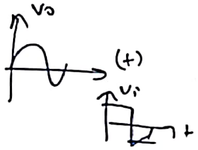
$V_i < V_r \Rightarrow V_x = +V_{cc}$

* $V_x = -V_{cc}$ ise;

$V_r = -\dfrac{V_{cc}}{2}$

Trigger (Tetikleyici)

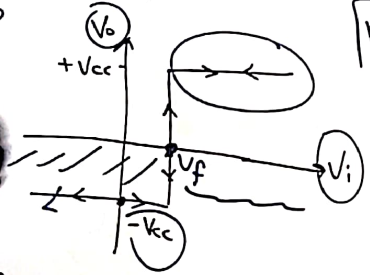
Sembolü



$V_i > V_f$ ise $+V_{cc}$

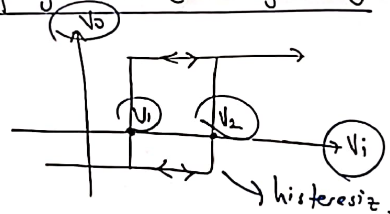
$V_i < V_f$ ise $-V_{cc}$

$\boxed{V_o = f(V_i)}$

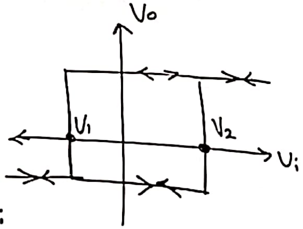


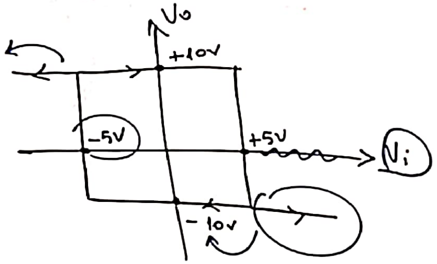
... de gürültü engellemek için kullanılır.

V_f gerilimi $[V_1 - V_2]$ aralığında değişirse;

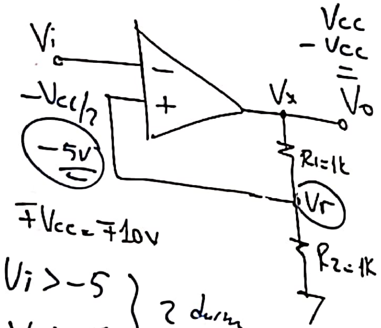


histerezis eğrisi





Örnek:



$\mp V_{cc} = \mp 10V$

$\left.\begin{array}{l} V_i > -5 \\ V_i < -5 \end{array}\right\}$ 2 durum

$V_r = V_x \cdot \frac{R_2}{R_1 + R_2}$

* $V_x = +V_{cc}$ ise;

$+5V \quad V_r = V_{cc} \cdot \frac{1k}{2k} = \frac{V_{cc}}{2}$

$V_i > V_r \Rightarrow V_x = -V_{cc}$

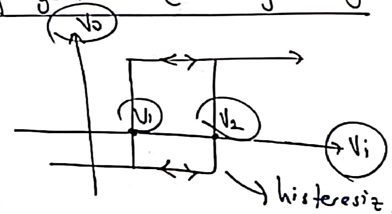
$-5V \quad V_i < V_r \Rightarrow V_x = +V_{cc}$

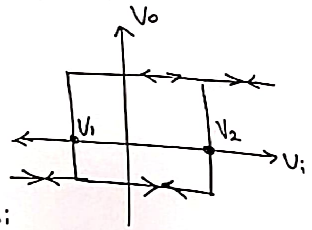
* $V_x = -V_{cc}$ ise;

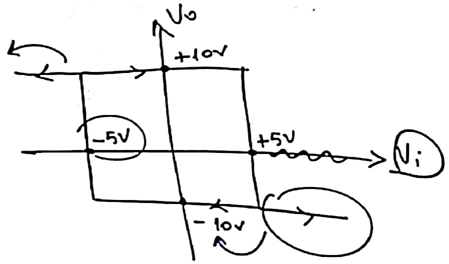
$V_r = -\frac{V_{cc}}{2}$

(Tetikleyici)

Vf gerilimi [V1–V2] aralığında değişirse;

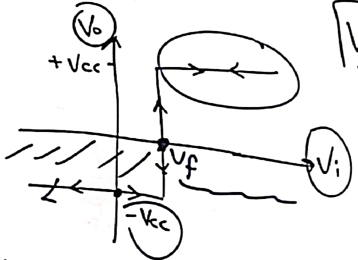






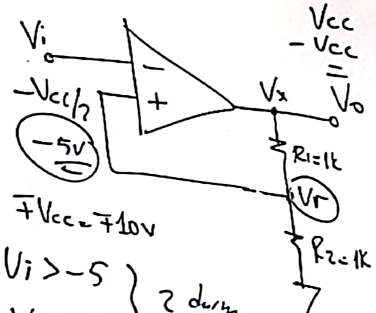
$V_i > V_f$ ise $+V_{cc}$

$V_i < V_f$ ise $-V_{cc}$



$\boxed{V_o = f(V_i)}$

Örnek:



$\mp V_{cc} = \mp 10v$

$\left.\begin{array}{l} V_i > -5 \\ V_i < -5 \end{array}\right\}$ 2 durum

$V_r = V_x \cdot \frac{R_2}{R_1+R_2}$

* $V_x = +V_{cc}$ ise;

+5V $\quad V_r = V_{cc} \cdot \frac{1k}{2k} = \frac{V_{cc}}{2}$

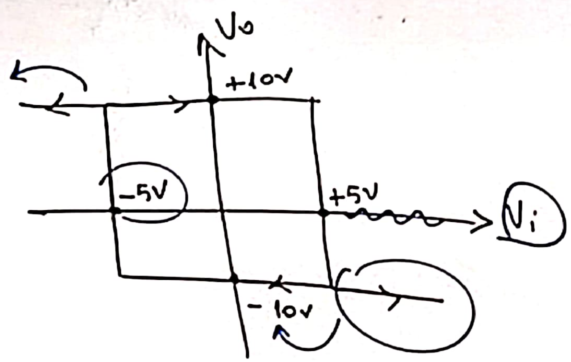
$V_i > V_r \Rightarrow V_x = -V_{cc}$

−5V $\quad V_i < V_r \Rightarrow V_x = +V_{cc}$

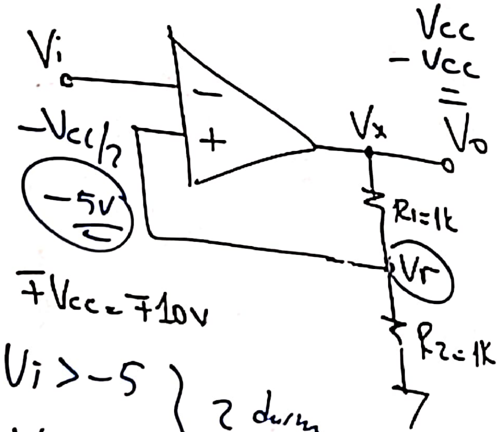
* $V_x = -V_{cc}$ ise;

$V_r = -\frac{V_{cc}}{2}$

...devreler-de gürültü engellemek için kullanılır.



Örnek:



$\mp V_{cc} = \mp 10V$

$V_i > -5$
$V_i < -5$ } 2 durum

$$V_r = V_x \cdot \frac{R_2}{R_1 + R_2}$$

* $V_x = +V_{cc}$ ise;

$+5V$ $$V_r = V_{cc} \cdot \frac{1k}{2k} = \frac{V_{cc}}{2}$$

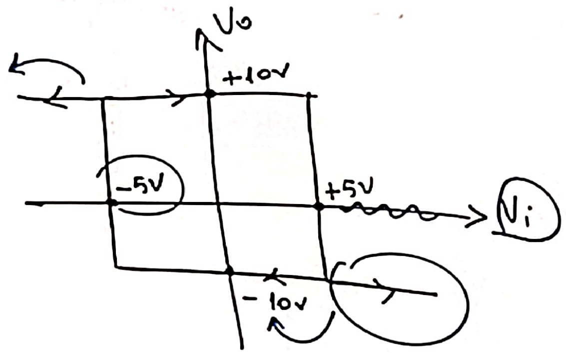
$V_i > V_r \Rightarrow V_x = -V_{cc}$

$-5V$

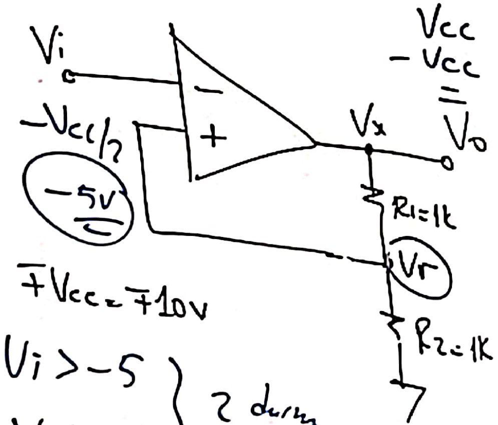
$V_i < V_r \Rightarrow V_x = +V_{cc}$

* $V_x = -V_{cc}$ ise;

$$V_r = -\frac{V_{cc}}{2}$$



Örnek:



$\mp V_{cc} = \mp 10V$

$\left.\begin{array}{l} V_i > -5 \\ V_i < -5 \end{array}\right\}$ 2 durum

$V_r = V_x \cdot \dfrac{R_2}{R_1+R_2}$

* $V_x = +V_{cc}$ ise;

$+5V$ $V_r = V_{cc} \cdot \dfrac{1k}{2k} = \dfrac{V_{cc}}{2}$

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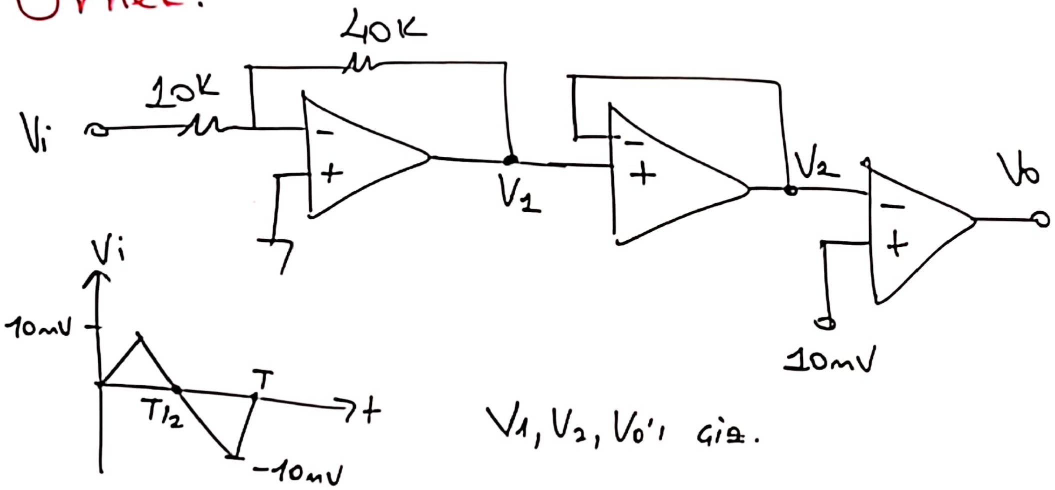
$V_r = -\dfrac{V_{cc}}{2}$

Örnek:

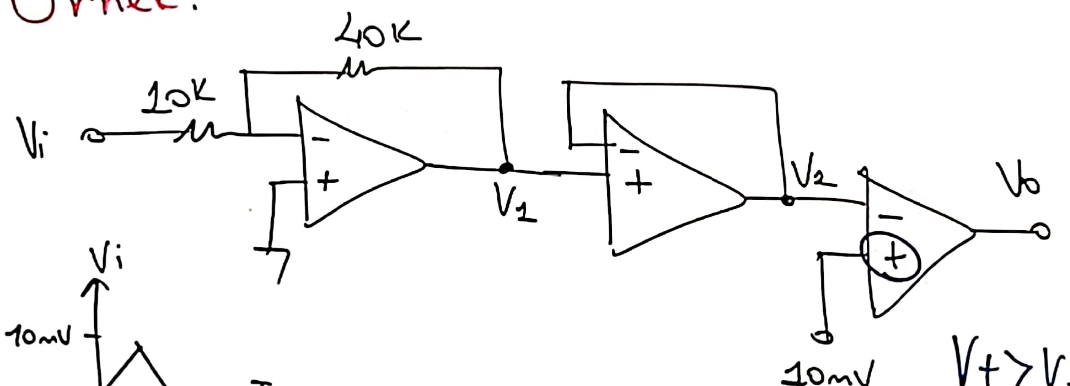
$\pm V_{cc} = \pm 10V$

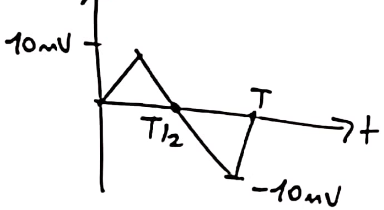


V_1, V_2, V_0'ı çiz.

Örnek:

$\pm V_{cc} = \pm 10V$

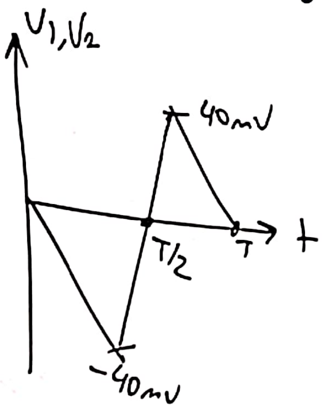




V_1, V_2, V_o'ı çiz.

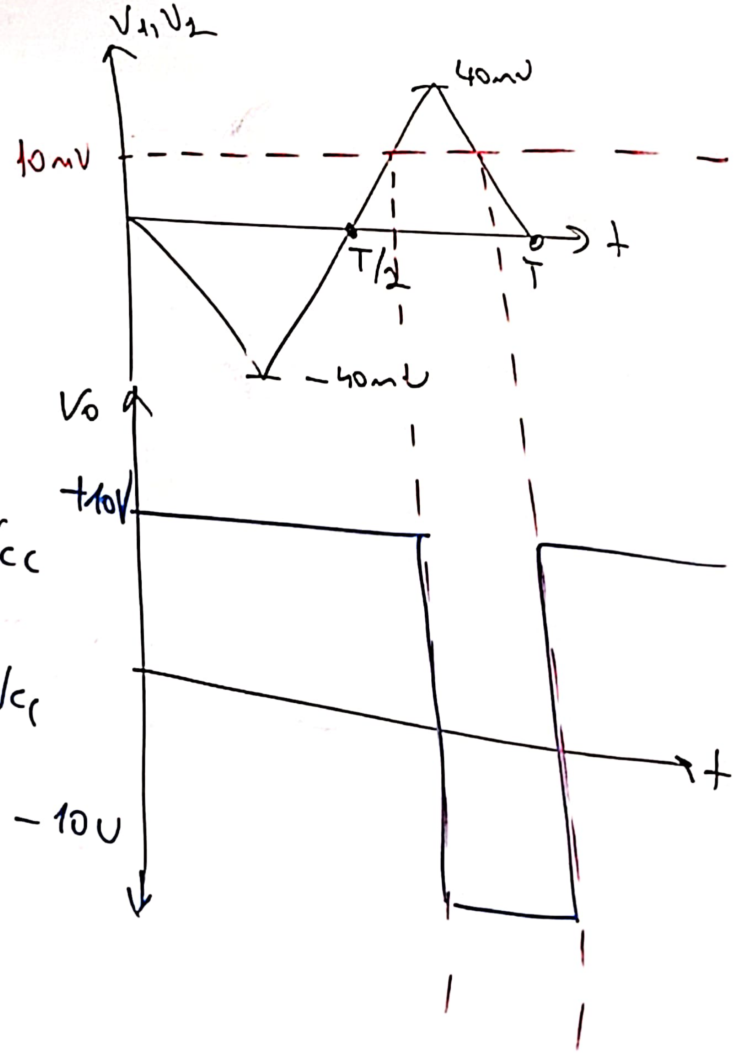
$$V_1 = -\left(\frac{40K}{10K}\right) V_i = -4 \cdot V_i$$

$$\boxed{V_2 = V_1}$$

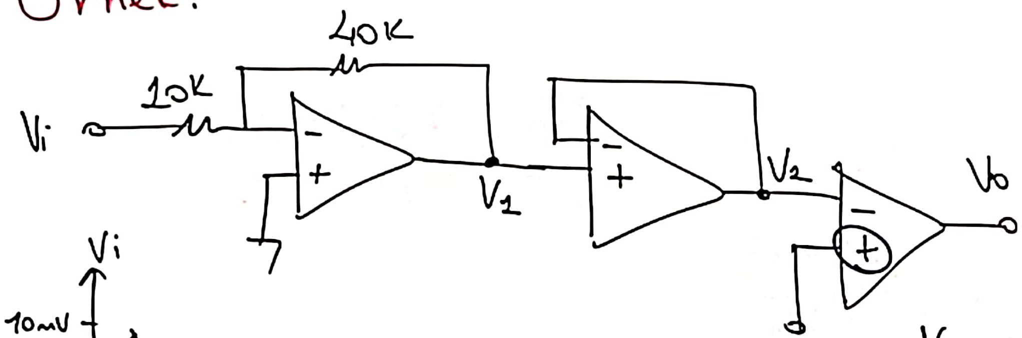


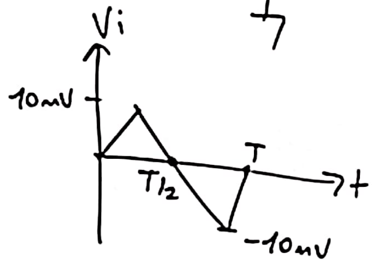
$V_+ > V_-$
$\Rightarrow +V_{cc}$

$V_- > V_+$
$\Rightarrow -V_{cc}$



Örnek: $\pm V_{cc} = \pm 10V$





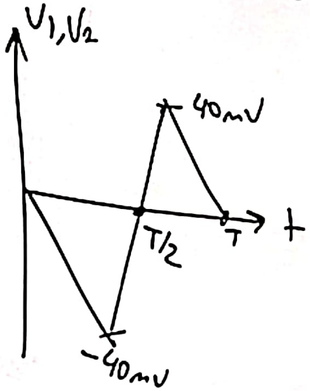
V_1, V_2, V_o'ı çiz.

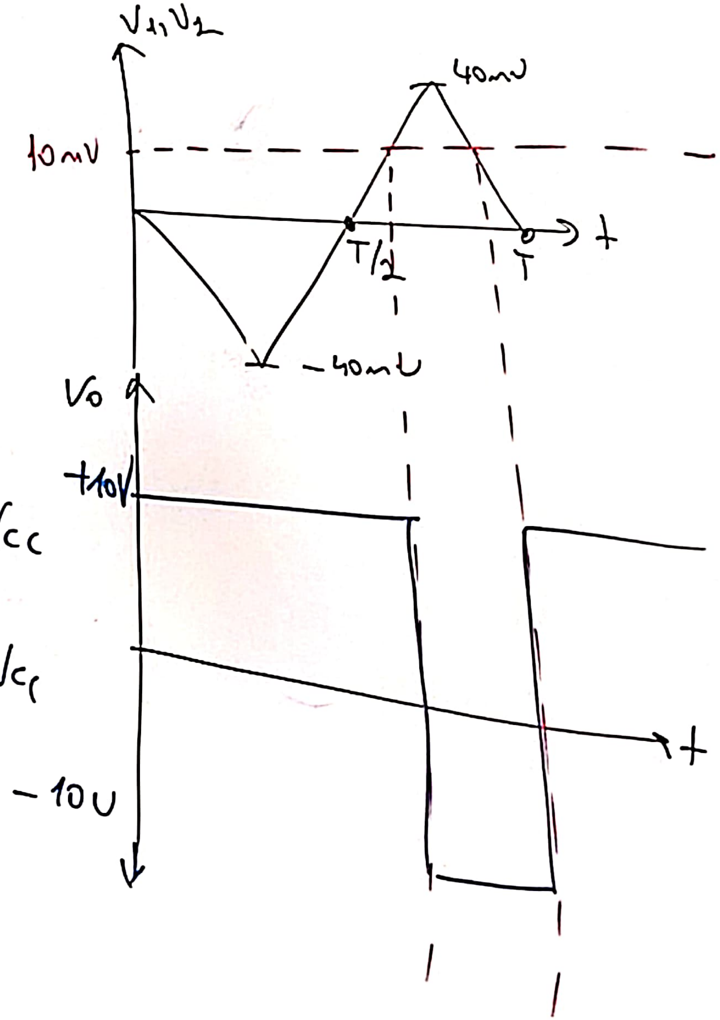
$V_+ > V_- \Rightarrow +V_{cc}$

$V_- > V_+ \Rightarrow -V_{cc}$

$$V_1 = -\left(\frac{40K}{10K}\right)V_i = -4.V_i$$

$\boxed{V_2 = V_1}$

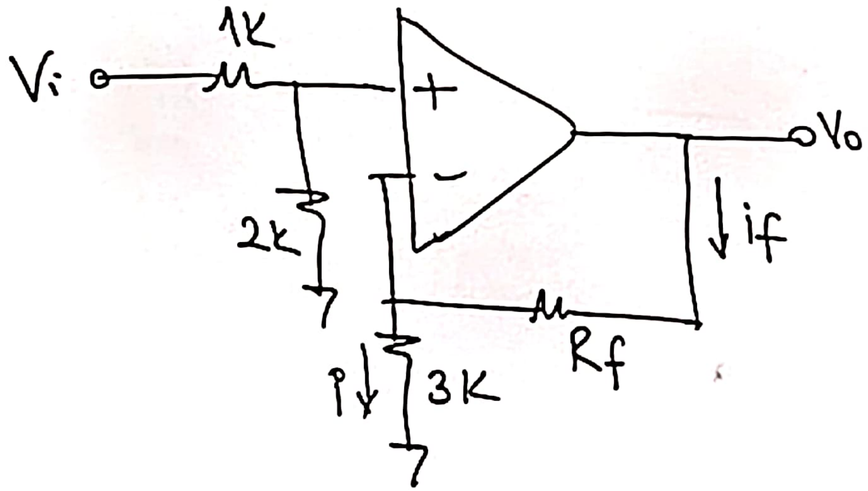




Sınav

Sınav Sorusu:

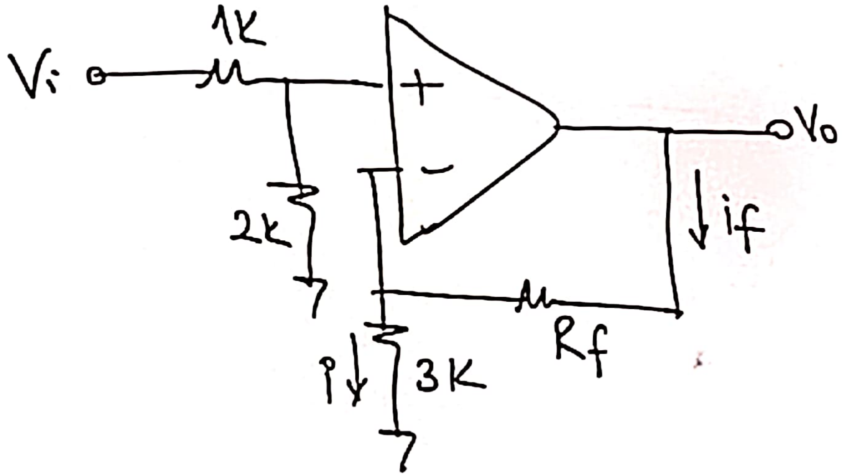
Aşağıda verilen ideal opamp'ın kazancının 5 olması için $R_f = ?$



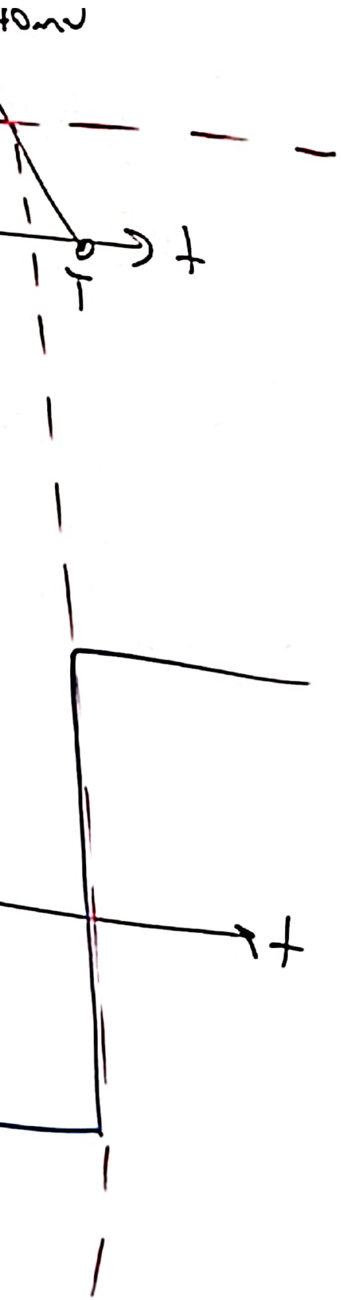
$$\frac{V_o}{V_i} = 5$$

Sınav Sorusu:

Aşağıda verilen ideal opamp'ın kazancının 5 olması için $R_f = ?$

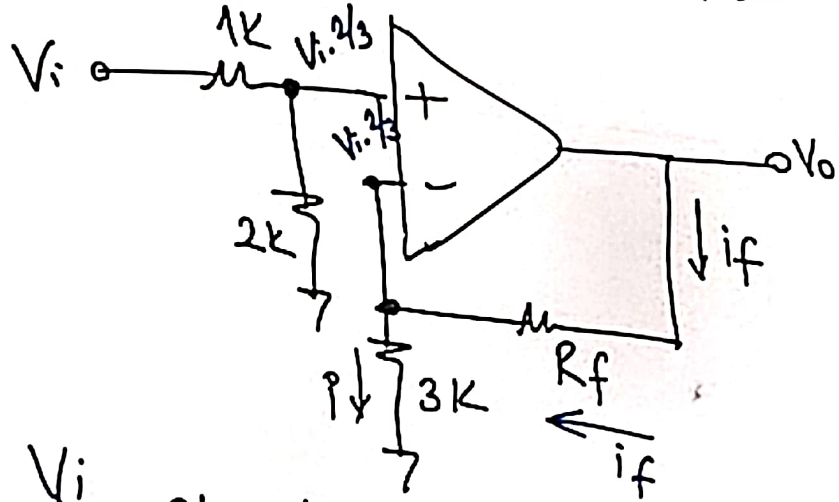


$$\frac{V_o}{V_i} = 5$$



Sınav Sorusu;

Aşağıda verilen ideal opamp'ın kazancının 5 olması için $R_f = ?$



$$\boxed{\frac{V_o}{V_i} = 5} \qquad V_o = 5V_i$$

$$V_+ = V_-$$
$$i_+ = i_- = 0$$

$$V_+ = \frac{V_i}{(2k+1k)} \cdot 2k = V_i \cdot \frac{2}{3}$$

$$i_f = i$$

$$\frac{V_o - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$$

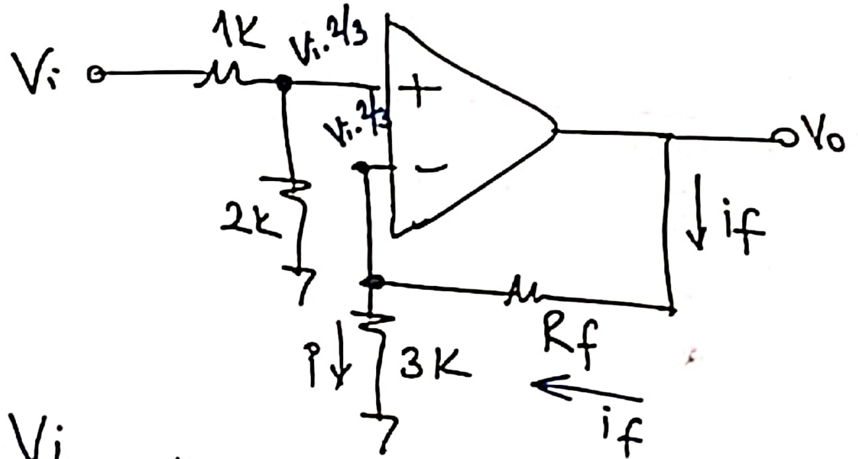
$$\frac{5V_i - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$$

$$\frac{\frac{13\cancel{V_i}}{\cancel{3}}}{R_f} = \frac{\frac{2}{\cancel{3}}\cancel{V_i}}{3k} \quad R_f$$

$$\frac{39}{2} = R_f \Rightarrow \boxed{R_f = 19.5k}$$

Sınav Sorusu:

Aşağıda verilen ideal opampın kazancının 5 olması için $R_f = ?$



$$\boxed{\frac{V_o}{V_i} = 5} \qquad V_o = 5V_i$$

$$V_+ = V_-$$

$$i_+ = i_- = 0$$

$$V_+ = \frac{V_i}{(2k+1k)} \cdot 2k = V_i \cdot \frac{2}{3}$$

$$i_f = i$$

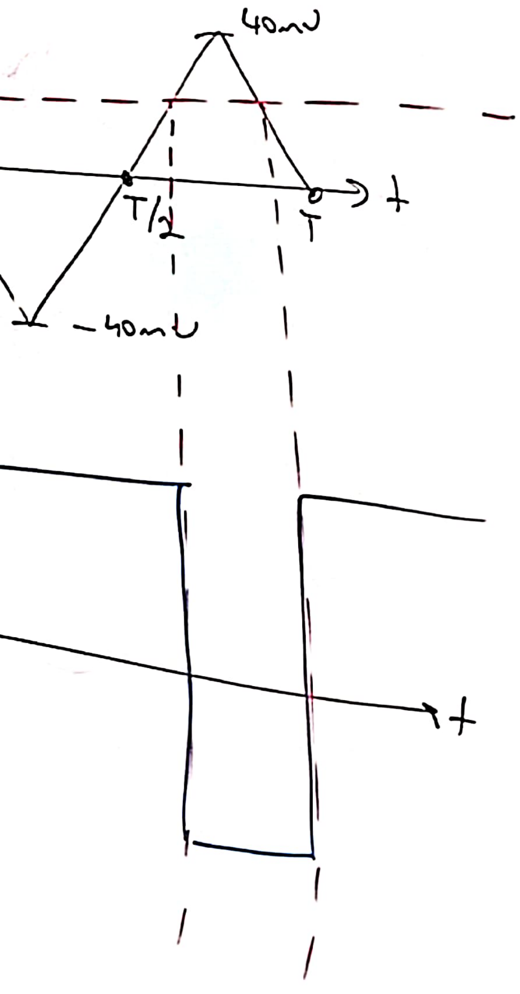
$$\downarrow$$

$$\frac{V_o - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$$

$$\frac{5V_i - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$$

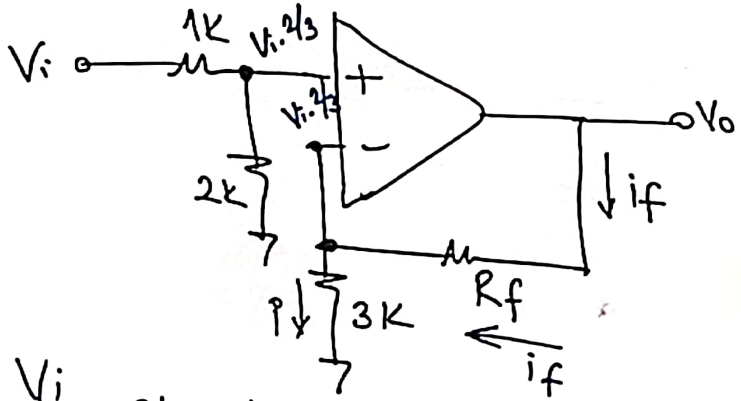
$$\frac{\frac{13\cancel{V_i}}{\cancel{3}}}{R_f} = \frac{\frac{2\cancel{V_i}}{\cancel{3}}}{3k}$$

$$\frac{39}{2} = R_f \Rightarrow \boxed{R_f = 19.5k}$$



Sınav Sorusu:

Aşağıda verilen ideal opamp'ın kazancının 5 olması için $R_f = ?$



$$\boxed{\frac{V_o}{V_i} = 5}$$

$V_o = 5V_i$

$V_+ = V_-$

$i_+ = i_- = 0$

$$V_+ = \frac{V_i}{(2k+1k)} \cdot 2k = V_i \cdot \frac{2}{3}$$

$$i_f = i$$

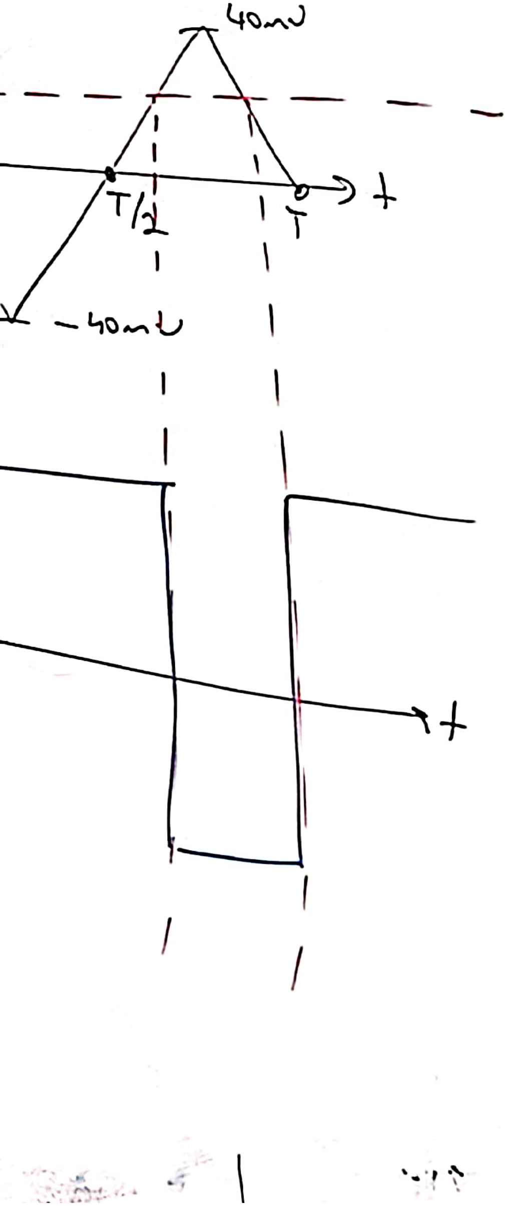
$$\downarrow$$

$$\frac{V_o - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$$

$$\frac{5V_i - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$$

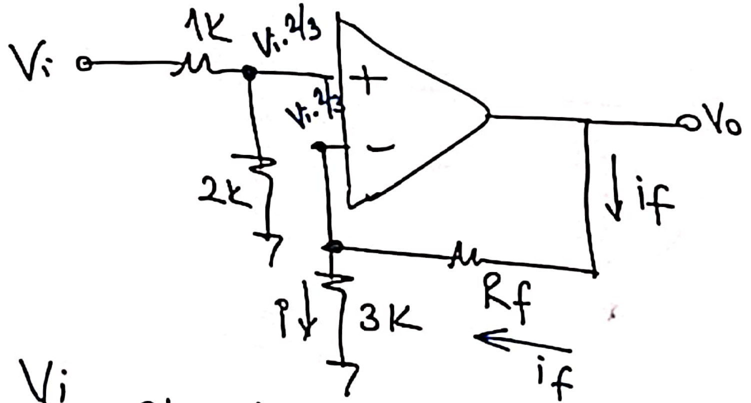
$$\frac{\frac{13\cancel{V_i}}{3}}{R_f} = \frac{\frac{2}{3}\cancel{V_i}}{3k}$$

$$\frac{39}{2} = R_f \Rightarrow \boxed{R_f = 19.5K}$$



Sınav Sorusu:

Aşağıda verilen ideal opamp'ın kazancının 5 olması için $R_f = ?$



$$\boxed{\frac{V_o}{V_i} = 5} \qquad V_o = 5V_i$$

$$V_+ = V_-$$

$$i_+ = i_- = 0$$

$$V_+ = \frac{V_i}{(2k+1k)} \cdot 2k = V_i \cdot \frac{2}{3}$$

$$i_f = i$$

$$\downarrow$$

$$\frac{V_o - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$$

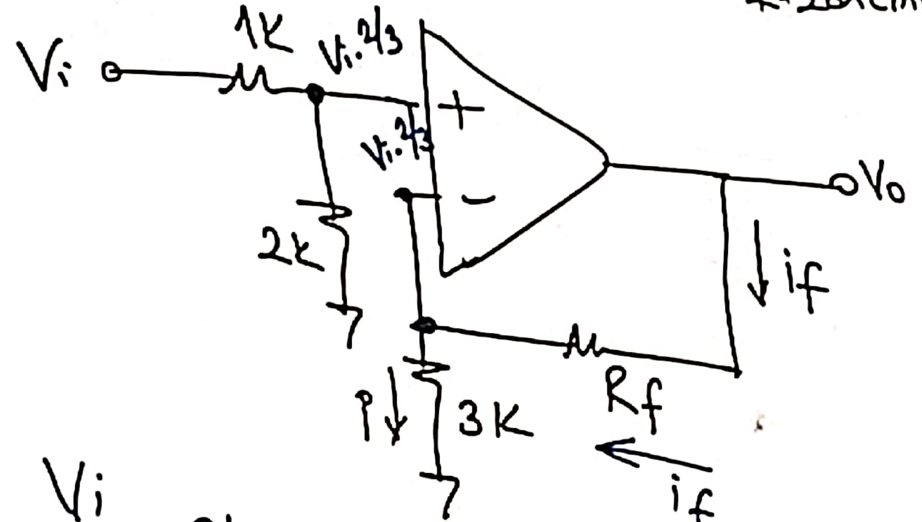
$$\frac{5V_i - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$$

$$\frac{\frac{13\cancel{V_i}}{\cancel{3}}}{R_f} = \frac{\frac{2}{\cancel{3}}\cancel{V_i}}{3k}$$

$$\frac{39}{2} = R_f \Rightarrow \boxed{R_f = 19.5k}$$

Sınav Sorusu:

Aşağıda verilen ideal opampın kazancının 5 olması için $R_f = ?$



$$\boxed{\frac{V_o}{V_i} = 5} \qquad V_o = 5V_i$$

$V_+ = V_-$

$i_+ = i_- = 0$

$$V_+ = \frac{V_i}{(2k+1k)} \cdot 2k = V_i \cdot \frac{2}{3}$$

$$i_f = i$$

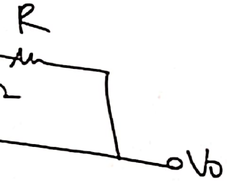
$$\downarrow$$

$$\frac{V_o - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$$

$$\frac{5V_i - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$$

$$\frac{\frac{13\cancel{V_i}}{\cancel{3}}}{R_f} = \frac{\frac{2}{\cancel{3}}\cancel{V_i}}{3k}$$

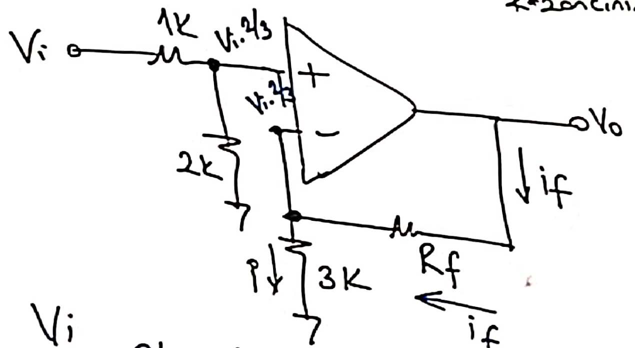
$$\frac{39}{2} = R_f \Rightarrow \boxed{R_f = 19.5K}$$



$\Rightarrow R = ?$

Sınav Sorusu:

Aşağıda verilen ideal opampın kazancının 5 olması için $R_f = ?$



$$\boxed{\frac{V_o}{V_i} = 5} \qquad V_o = 5V_i$$

$$V_+ = V_-$$
$$i_+ = i_- = 0$$

$$V_+ = \frac{V_i}{(2k+1k)} \cdot 2k = V_i \cdot \frac{2}{3}$$

$$i_f = i$$

$$\downarrow$$

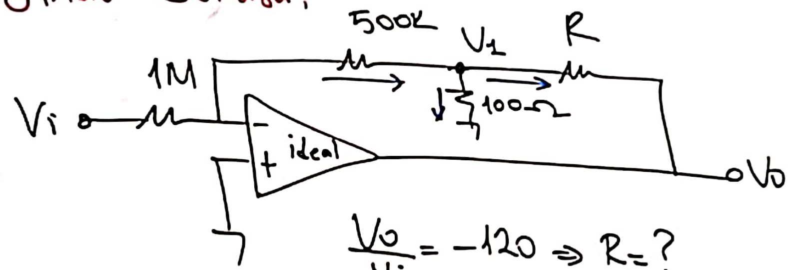
$$\frac{V_o - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$$

$$\frac{5V_i - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$$

$$\frac{\frac{13\cancel{V_i}}{3}}{R_f} = \frac{\frac{2}{3}\cancel{V_i}}{3k}$$

$$\frac{39}{2} = R_f \Rightarrow \boxed{R_f = 19.5k}$$

Sınav Sorusu;



$$\frac{V_o}{V_i} = -120 \Rightarrow R = ?$$

$$\frac{V_i - 0}{\cancel{1M}\,2} = \frac{0 - V_1}{5\cancel{00}K}$$

$$\boxed{V_i = -2V_1}$$

$$V_1 = -\frac{V_i}{2}$$

$$\frac{0 - V_1}{500k} = \frac{V_1 - 0}{100\Omega} + \frac{V_1 - V_o}{R}$$

$$\frac{-V_1}{500k} = \frac{V_1}{100\Omega} + \frac{V_1 - V_o}{R}$$

$$\frac{\frac{V_i}{2}}{500K} = \frac{\frac{-V_i}{2}}{100\Omega} + \frac{\left(-\frac{V_i}{2}\right) - V_o}{R}$$

$$V_o = -120 V_i \Rightarrow R = ?$$

Sınav Sor

V_i

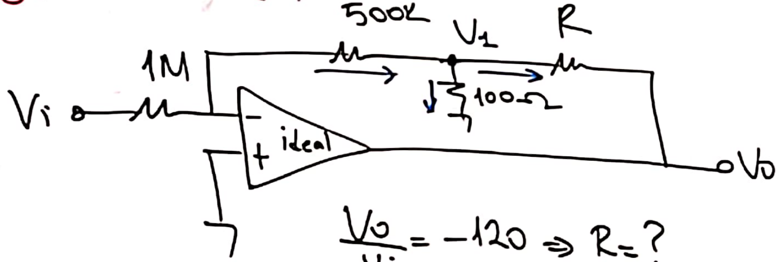
$$V_+ = \frac{V_i}{(2k+1k)}$$

$$\frac{13 \cancel{V_i}}{\cancel{3}}$$

Rf

$$\frac{39}{2} = Rf \Rightarrow Rf =$$

Sınav Sorusu:



$$\frac{V_0}{V_i} = -120 \Rightarrow R = ?$$

$$\frac{V_i - 0}{1\cancel{M}_2} = \frac{0 - V_1}{5\cancel{0}0K}$$

$$\boxed{V_i = -2V_1}$$

$$V_1 = -\frac{V_i}{2}$$

$$\frac{0 - V_1}{500k} = \frac{V_1 - 0}{100\Omega} + \frac{V_1 - V_0}{R}$$

$$\frac{-V_1}{500K} = \frac{V_1}{100\Omega} + \frac{V_1 - V_0}{R}$$

$$\frac{\frac{V_i}{2}}{500K} = \frac{-\frac{V_i}{2}}{100\Omega} + \frac{\left(-\frac{V_i}{2}\right) - V_0}{R}$$

$$V_0 = -120\,V_i \Rightarrow R = ?$$

Sınav Sor

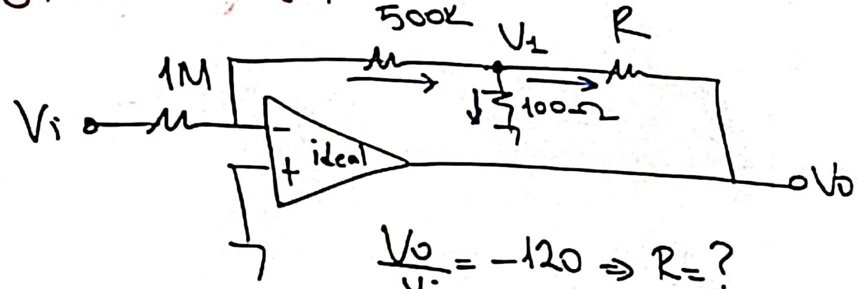
V_i o—

$$V_+ = \frac{V_i}{(2k+1k)}$$

$$\frac{\frac{13\cancel{V_i}}{\cancel{3}}}{Rf}$$

$$\frac{39}{2} = Rf \Rightarrow \boxed{Rf = }$$

Sınav Sorusu;



$\frac{V_o}{V_i} = -120 \Rightarrow R = ?$

$\frac{V_i - 0}{\cancel{1M}_{2}} = \frac{0 - V_1}{5\cancel{00}K}$

$\boxed{V_i = -2V_1}$

$V_1 = -\frac{V_i}{2}$

$\frac{0 - V_1}{500k} = \frac{V_1 - 0}{100\Omega} + \frac{V_1 - V_o}{R}$

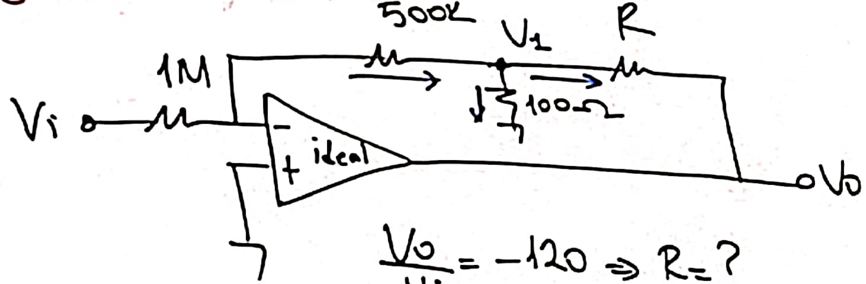
$\frac{-V_1}{500k} = \frac{V_1}{100\Omega} + \frac{V_1 - V_o}{R}$

$\frac{\frac{V_i}{2}}{500K} = \frac{\frac{-V_i}{2}}{100\Omega} + \frac{\left(-\frac{V_i}{2}\right) - V_o}{R}$

$V_o = -120 V_i \Rightarrow R = ?$

$\frac{3}{2}$

Sınav Sorusu:



$$\frac{V_o}{V_i} = -120 \Rightarrow R = ?$$

$$\frac{V_i - 0}{\cancel{1M}\,2} = \frac{0 - V_1}{5\cancel{0}0K}$$

$$\boxed{V_i = -2V_1}$$

$$V_1 = -\frac{V_i}{2}$$

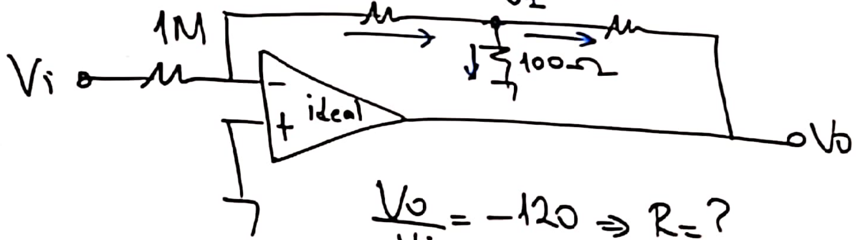
$$\frac{0 - V_1}{500k} = \frac{V_1 - 0}{100\Omega} + \frac{V_1 - V_o}{R}$$

$$\frac{-V_1}{500k} = \frac{V_1}{100\Omega} + \frac{V_1 - V_o}{R}$$

$$\frac{\frac{V_i}{2}}{500k} = \frac{\frac{-V_i}{2}}{100\Omega} + \frac{\left(-\frac{V_i}{2}\right) - V_o}{R}$$

$$V_o = -120 V_i \Rightarrow R = ?$$

Sınav Sorusu:



$$\frac{V_o}{V_i} = -120 \Rightarrow R = ?$$

$$\frac{V_i - 0}{\cancel{1M}_2} = \frac{0 - V_1}{5\cancel{0}0K}$$

$$\boxed{V_i = -2V_1}$$

$$V_1 = -\frac{V_i}{2}$$

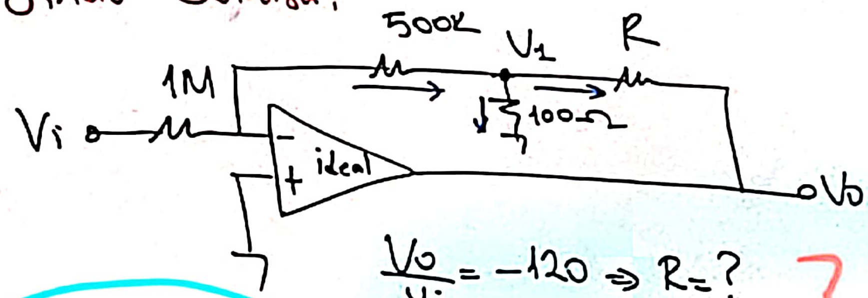
$$\frac{0 - V_1}{500k} = \frac{V_1 - 0}{100\Omega} + \frac{V_1 - V_o}{R}$$

$$\frac{-V_1}{500K} = \frac{V_1}{100\Omega} + \frac{V_1 - V_o}{R}$$

$$\frac{\frac{V_i}{2}}{500K} = \frac{\frac{-V_i}{2}}{100\Omega} + \frac{\left(-\frac{V_i}{2}\right) - V_o}{R}$$

$$V_o = -120 V_i \Rightarrow R = ?$$

Sınav Sorusu:



$$\frac{V_0}{V_i} = -120 \Rightarrow R = ?$$

2

1

$$\frac{V_i - 0}{\cancel{1M}_2} = \frac{0 - V_1}{\cancel{500}K}$$

$$\boxed{V_i = -2V_1}$$

$$V_1 = -\frac{V_i}{2}$$

$$\frac{0 - V_1}{500k} = \frac{V_1 - 0}{100\Omega} + \frac{V_1 - V_0}{R}$$

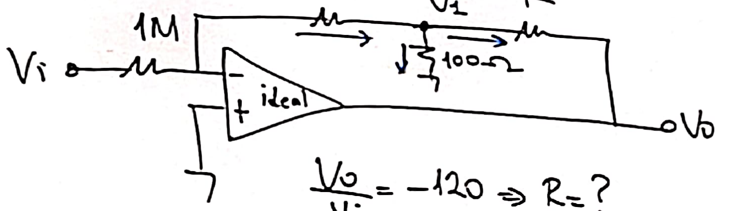
$$\frac{-V_1}{500k} = \frac{V_1}{100\Omega} + \frac{V_1 - V_0}{R}$$

$$\frac{\frac{V_i}{2}}{500k} = \frac{\frac{-V_i}{2}}{100\Omega} + \frac{\left(-\frac{V_i}{2}\right) - V_0}{R}$$

$$V_0 = -120 V_i \Rightarrow R = ?$$

3

Sınav Sorusu:



$\frac{V_o}{V_i} = -120 \Rightarrow R = ?$

$\frac{V_i - 0}{\cancel{1M}\,2} = \frac{0 - V_1}{5\cancel{00}K}$

$\boxed{V_i = -2V_1}$

$V_1 = -\frac{V_i}{2}$

$\frac{0 - V_1}{500k} = \frac{V_1 - 0}{100\Omega} + \frac{V_1 - V_0}{R}$

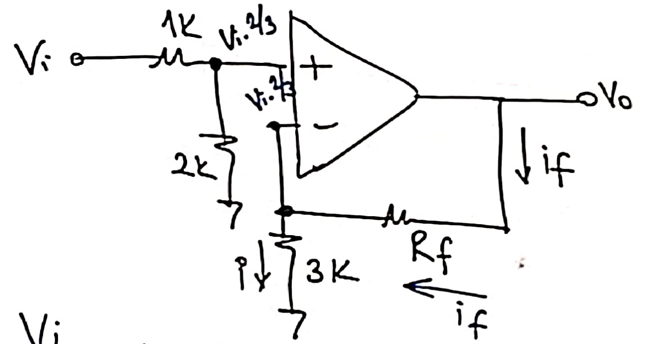
$\frac{-V_1}{500k} - \frac{V_1}{100\Omega} + \frac{V_1 - V_o}{R}$

$\frac{\frac{V_i}{2}}{500K} = \frac{\frac{-V_i}{2}}{100\Omega} + \frac{\left(-\frac{V_i}{2}\right) - V_o}{R}$

$V_o = -120V_i \Rightarrow R = ?$

Sınav Sorusu:

Aşağıda verilen ideal opamp'ın kazancının 5 olması için $R_f = ?$



$\boxed{\frac{V_o}{V_i} = 5}$ $\quad V_o = 5V_i$

$V_+ = V_-$

$i_+ = i_- = 0$

$V_+ = \frac{V_i}{(2k+1k)} \cdot 2k = V_i \cdot \frac{2}{3}$

$i_f = i$

$\downarrow$

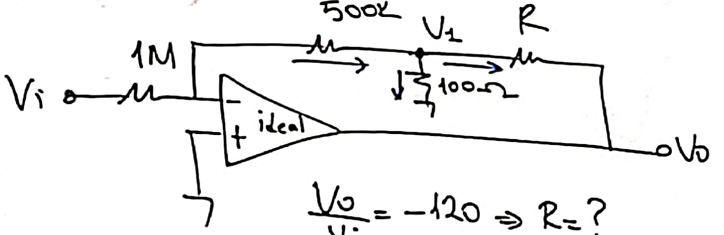
$\frac{V_o - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$

$\frac{5V_i - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$

$\frac{\frac{13\cancel{V_i}}{3}}{R_f} = \frac{\frac{2}{3}\cancel{V_i}}{3k}$

$\frac{39}{2} = R_f \Rightarrow \boxed{R_f = 19.5k}$

Sınav Sorusu:



$$\frac{V_o}{V_i} = -120 \Rightarrow R = ?$$

$$\frac{V_i - 0}{\cancel{1M}_{2}} = \frac{0 - V_1}{5\cancel{0}0K}$$

$$\boxed{V_i = -2V_1}$$

$$V_1 = -\frac{V_i}{2}$$

$$\frac{0 - V_1}{500k} = \frac{V_1 - 0}{100\Omega} + \frac{V_1 - V_o}{R}$$

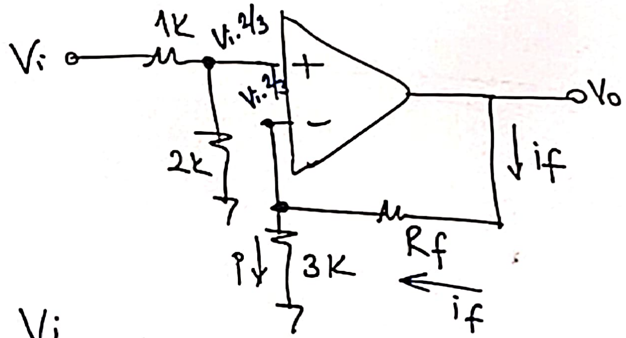
$$\frac{-V_1}{500k} = \frac{V_1}{100\Omega} + \frac{V_1 - V_o}{R}$$

$$\frac{\frac{V_i}{2}}{500k} = \frac{\frac{-V_i}{2}}{100\Omega} + \frac{\left(-\frac{V_i}{2}\right) - V_o}{R}$$

$V_o = -120 V_i \Rightarrow R = ?$

Sınav Sorusu:

Aşağıda verilen ideal opamp'ın kazancının 5 olması için $R_f = ?$



$$\boxed{\frac{V_o}{V_i} = 5}$$

$V_o = 5V_i$

$V_+ = V_-$

$i_+ = i_- = 0$

$$V_+ = \frac{V_i}{(2k + 1k)} \cdot 2k = V_i \cdot \frac{2}{3}$$

$i_f = i$

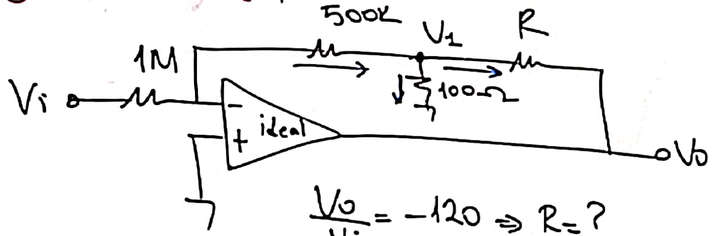
$$\frac{V_o - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$$

$$\frac{5V_i - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$$

$$\frac{\frac{13\cancel{V_i}}{\cancel{3}}}{R_f} = \frac{\frac{2}{\cancel{3}}\cancel{V_i}}{3k}$$

$$\frac{39}{2} = R_f \Rightarrow \boxed{R_f = 19.5k}$$

Sınav Sorusu:



$\frac{V_o}{V_i} = -120 \Rightarrow R = ?$

$\frac{V_i - 0}{\cancel{1M}\,2} = \frac{0 - V_1}{5\cancel{00}K}$

$\boxed{V_i = -2V_1}$

$V_1 = -\frac{V_i}{2}$

$\frac{0 - V_1}{500k} = \frac{V_1 - 0}{100\Omega} + \frac{V_1 - V_0}{R}$

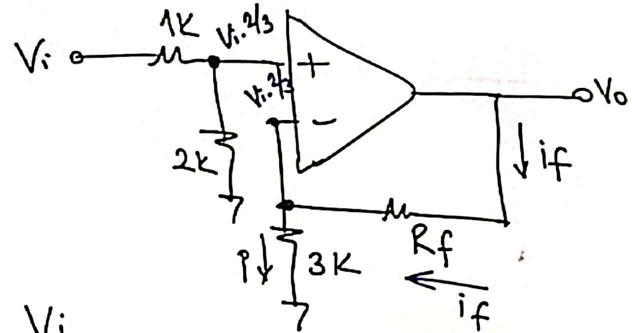
$\frac{-V_1}{500k} = \frac{V_1}{100\Omega} + \frac{V_1 - V_0}{R}$

$\frac{\frac{V_i}{2}}{500K} = \frac{\frac{-V_i}{2}}{100\Omega} + \frac{\left(-\frac{V_i}{2}\right) - V_0}{R}$

$V_o = -120 V_i \Rightarrow R = ?$

Sınav Sorusu:

Aşağıda verilen ideal opampın kazancının 5 olması için $R_f = ?$



$\boxed{\frac{V_o}{V_i} = 5}$ $\quad V_o = 5V_i$

$V_+ = V_-$

$i_+ = i_- = 0$

$V_+ = \frac{V_i}{(2k+1k)} \cdot 2k = V_i \cdot \frac{2}{3}$

$i_f = i$

$\frac{V_o - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$

$\frac{5V_i - \frac{2}{3}V_i}{R_f} = \frac{\frac{2}{3}V_i}{3K}$

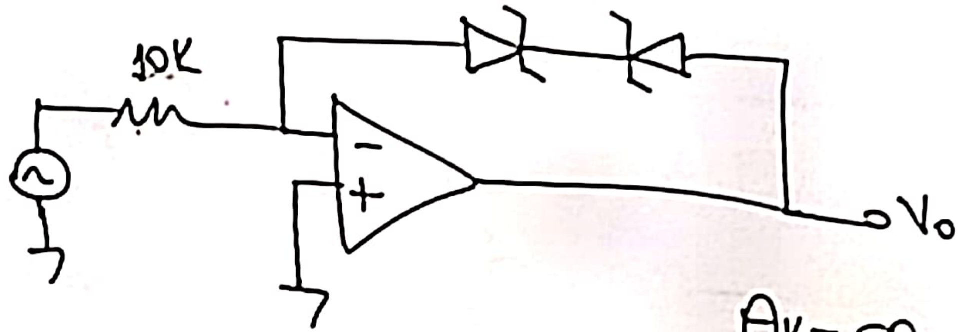
$\frac{\frac{13\cancel{V_i}}{\cancel{3}}}{R_f} = \frac{\frac{2}{\cancel{3}}\cancel{V_i}}{3k}$

$\frac{39}{2} = R_f \Rightarrow \boxed{R_f = 19.5k}$

Çalışma Sorusu

$V_{Z1} = V_{Z2} = 10V$ ($V_D = 0.7$)

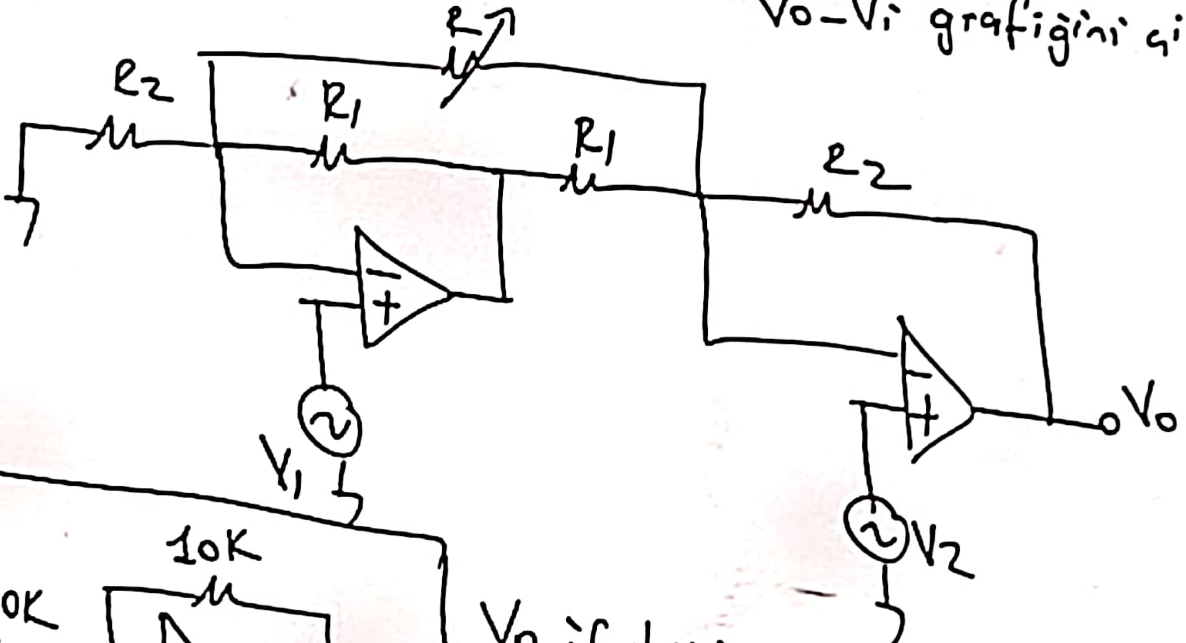
1.



$A_V = \infty$

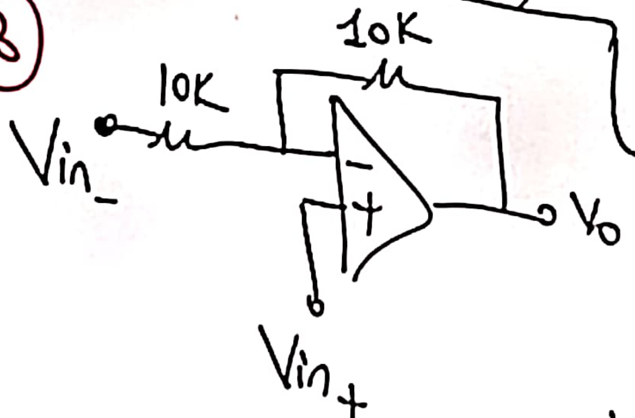
$V_o - V_i$ grafiğini çiz.

2.



V_o ifadesini elde et.

3.

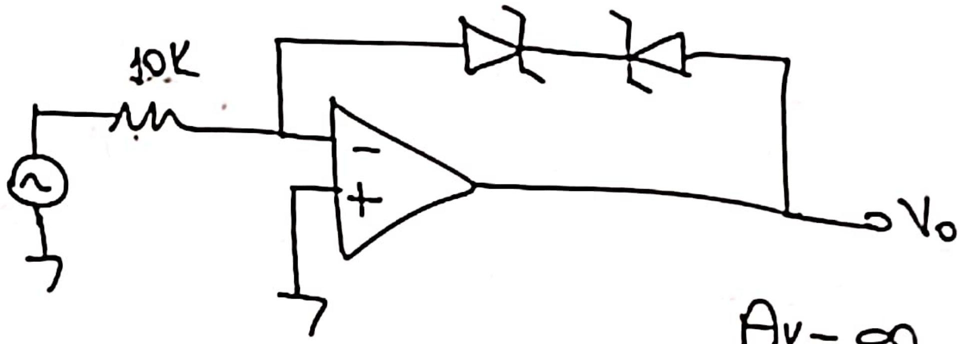


$V_{in\_} = 5V_{pp}$ Sin.

$\left.\begin{array}{l} V_{in+} = 5V \\ V_{in+} = -5V \end{array}\right\}$ olursa $V_o - V_{in\_}$ graf. çiz.

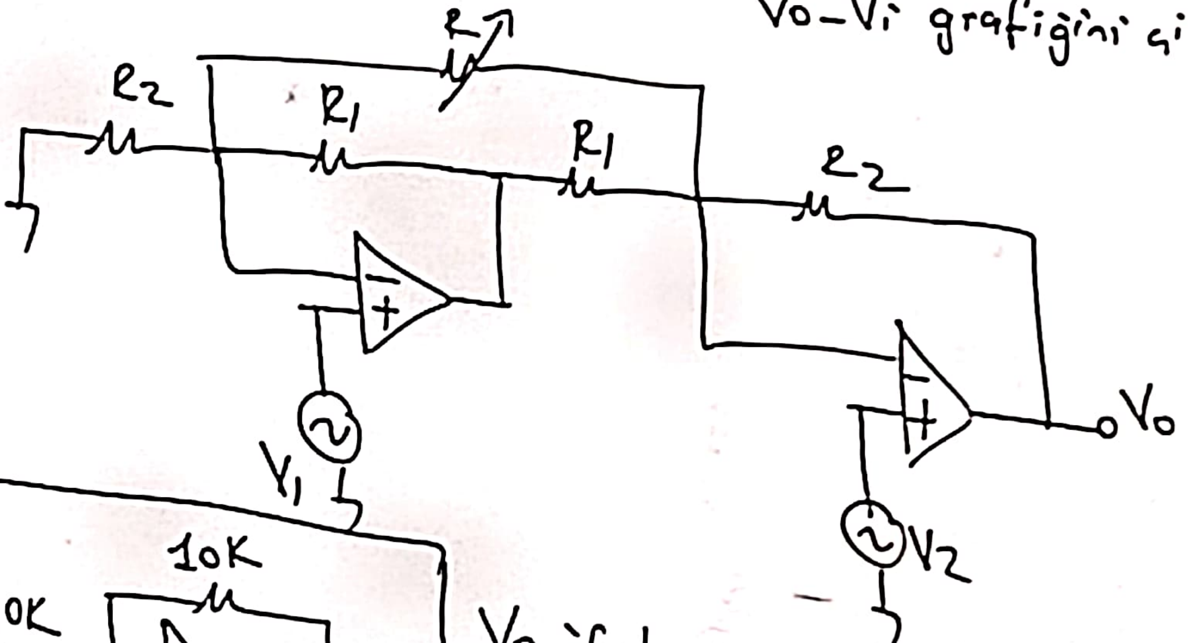
Çalışma Sorusu

$V_{Z1} = V_{Z2} = 10V$ $(V_D = 0.7)$

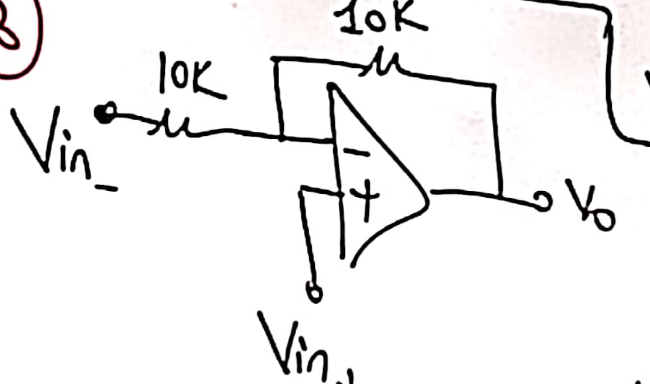
① 

$A_V = \infty$

$V_o - V_i$ grafiğini çiz.

② 

V_o ifadesini elde et.

③ 

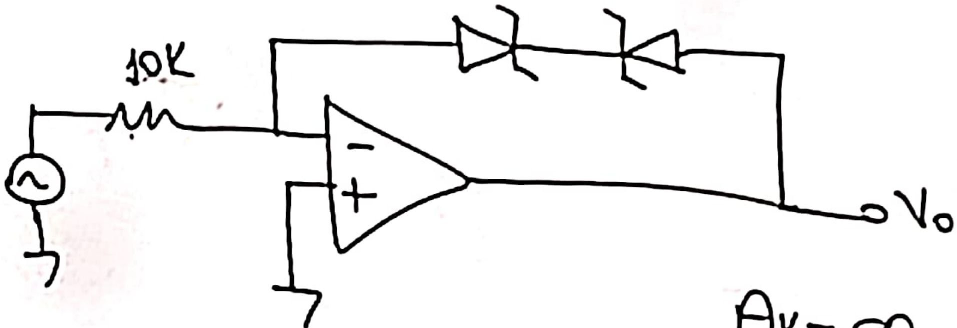
$V_{in-} = 5V_{pp}$ Sin.

$\left.\begin{array}{l} V_{in+} = 5V \\ V_{in+} = -5V \end{array}\right\}$ olursa $V_o - V_{in-}$ graf. çiz.

Çalışma Sorusu

$V_{z1} = V_{z2} = 10V$ $(V_D = 0.7)$

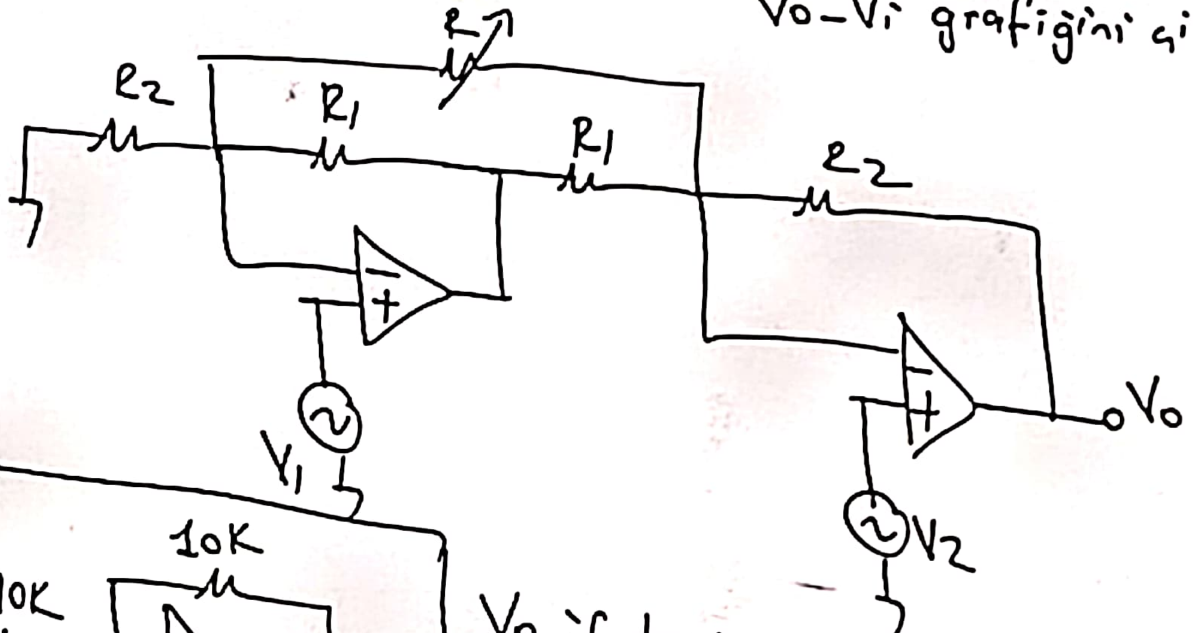
(1)



$A_v = \infty$

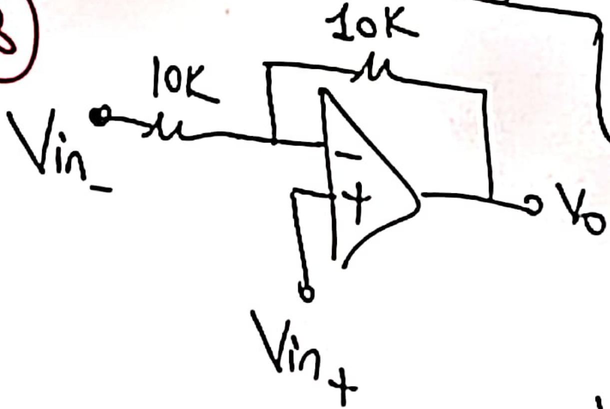
$V_o - V_i$ grafiğini çiz.

(2)



V_o ifadesini elde et.

(3)



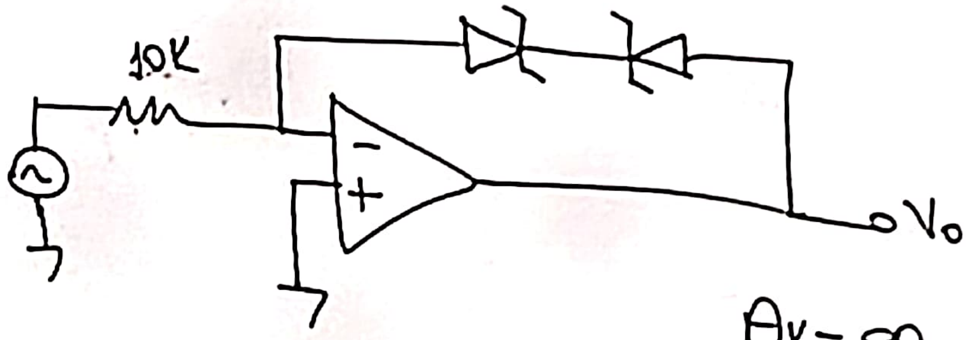
$V_{in-} = 5V_{pp}$ Sin.

$\left.\begin{array}{l} V_{in+} = 5V \\ V_{in+} = -5V \end{array}\right\}$ olursa $V_o - V_{in-}$ graf. çiz.

Çalışma Sorusu

$V_{Z1} = V_{Z2} = 10V$ $(V_D = 0.7)$

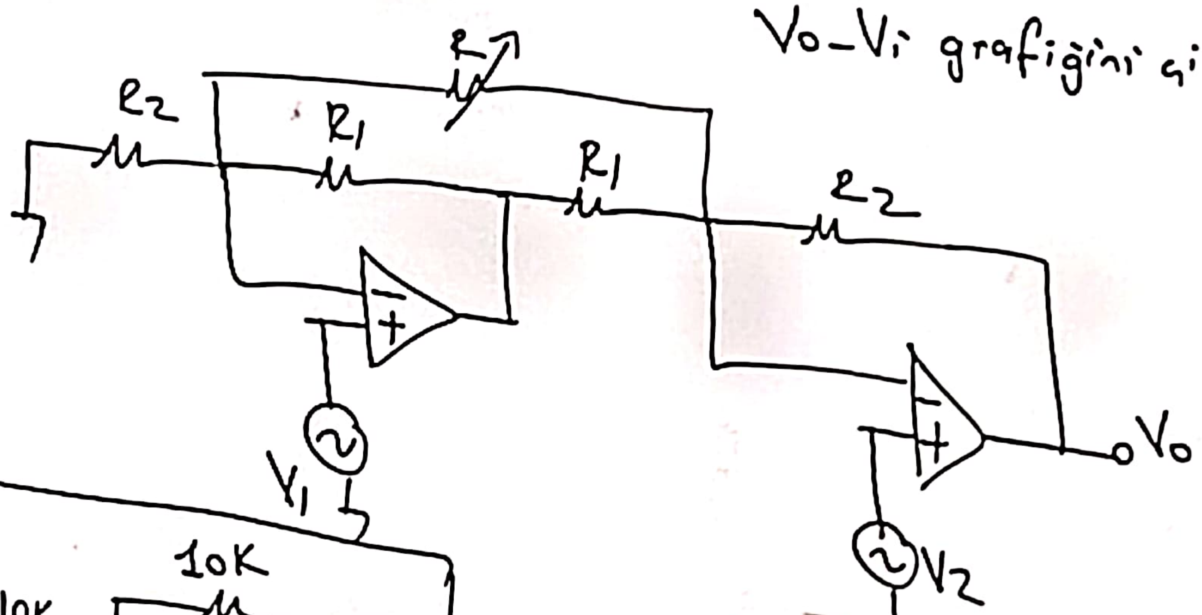
(1)



$A_V = \infty$

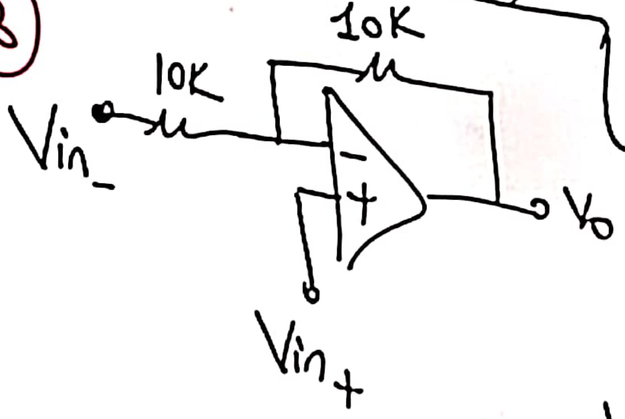
$V_o - V_i$ grafiğini çiz.

(2)



V_o ifadesini elde et.

(3)



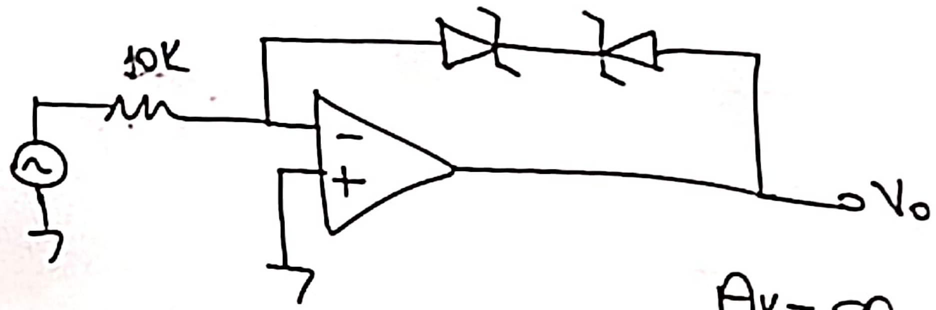
$V_{in\_} = 5V_{pp}$ Sin.

$\left.\begin{array}{l} V_{in+} = 5V \\ V_{in+} = -5V \end{array}\right\}$ olursa $V_o - V_{in\_}$ graf. çiz.

Çalışma Sorusu

$V_{Z1} = V_{Z2} = 10V$ ($V_D = 0.7$)

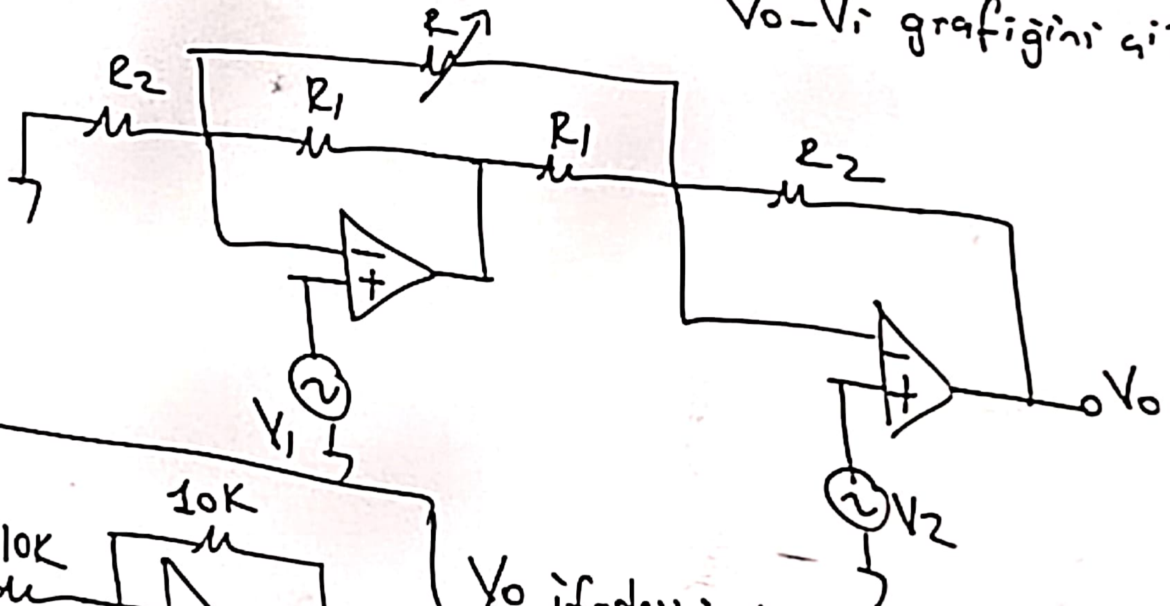
(1)



$A_V = \infty$

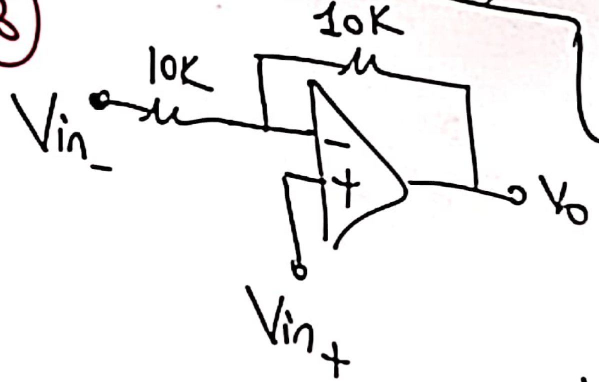
Vo–Vi grafiğini çiz.

(2)



Vo ifadesini elde et.

(3)



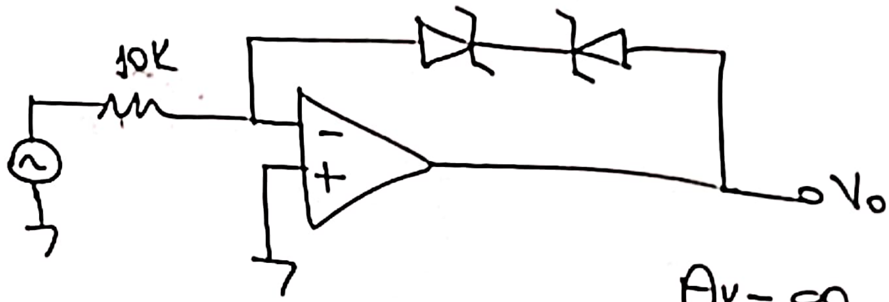
$V_{in\_} = 5V_{pp}$ Sin.

$V_{in+} = 5V$
$V_{in+} = -5V$
} olursa Vo – Vin_ graf. çiz.

Çalışma Sorusu

$V_{z1} = V_{z2} = 10V \quad (V_D = 0.7)$

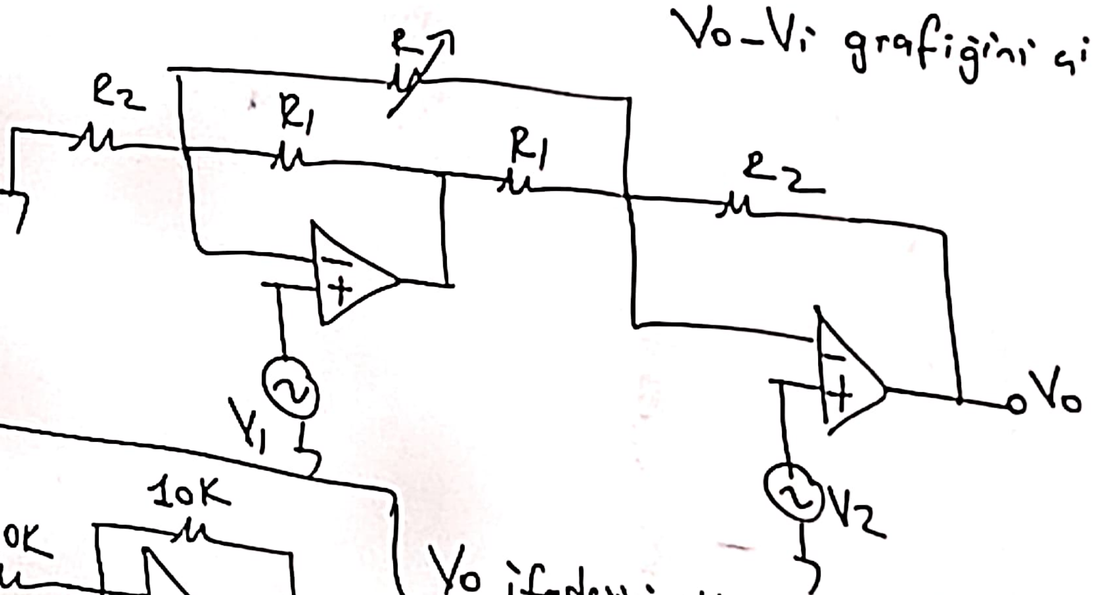
(1)



$A_v = \infty$

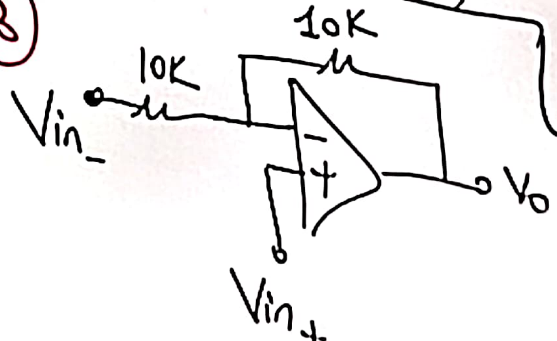
$V_o - V_i$ grafiğini çiz.

(2)



V_o ifadesini elde et.

(3)

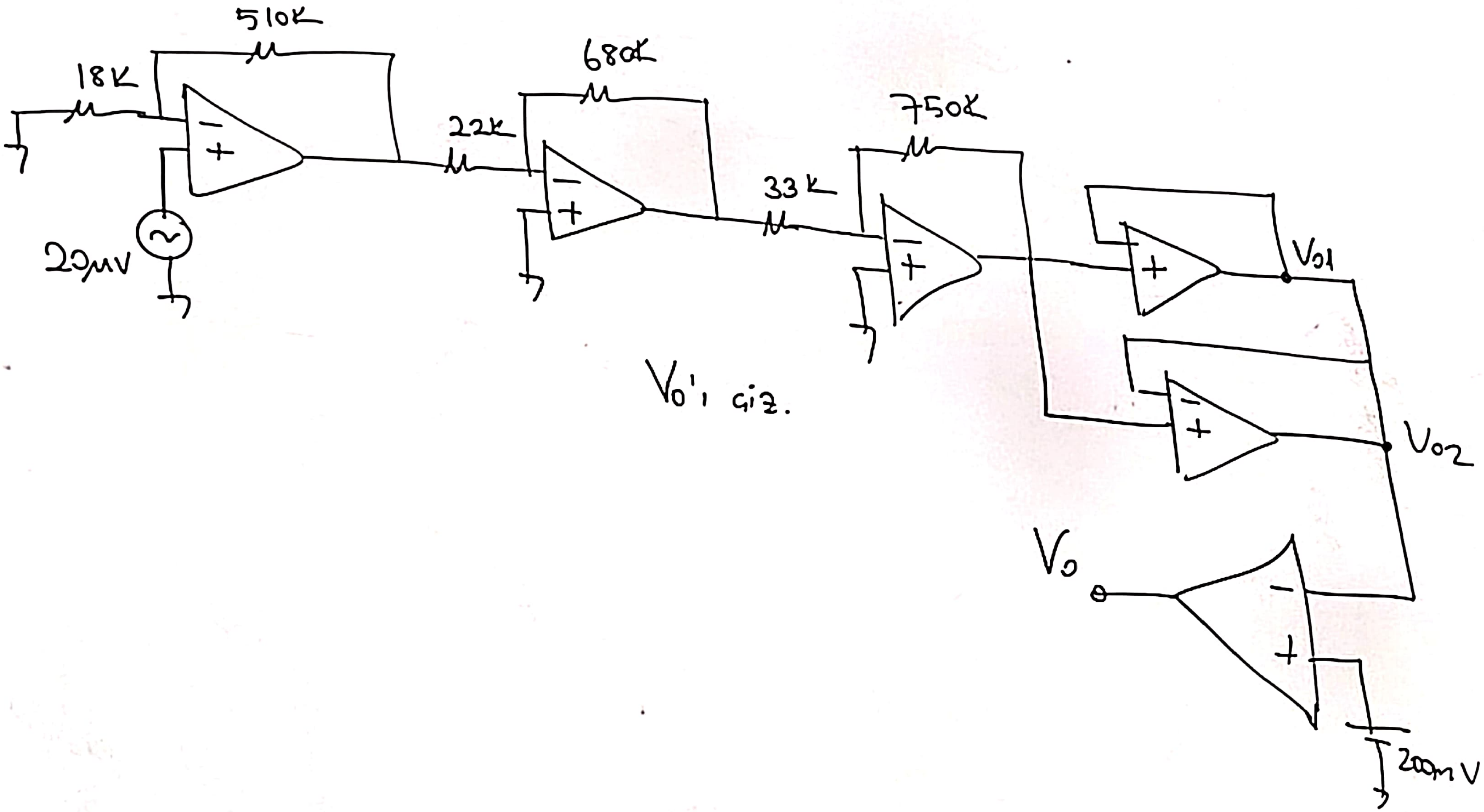


$V_{in-} = 5V_{pp}$ Sin.

$\left.\begin{array}{l} V_{in+} = 5V \\ V_{in+} = -5V \end{array}\right\}$ olursa $V_o - V_{in-}$ graf. çiz.

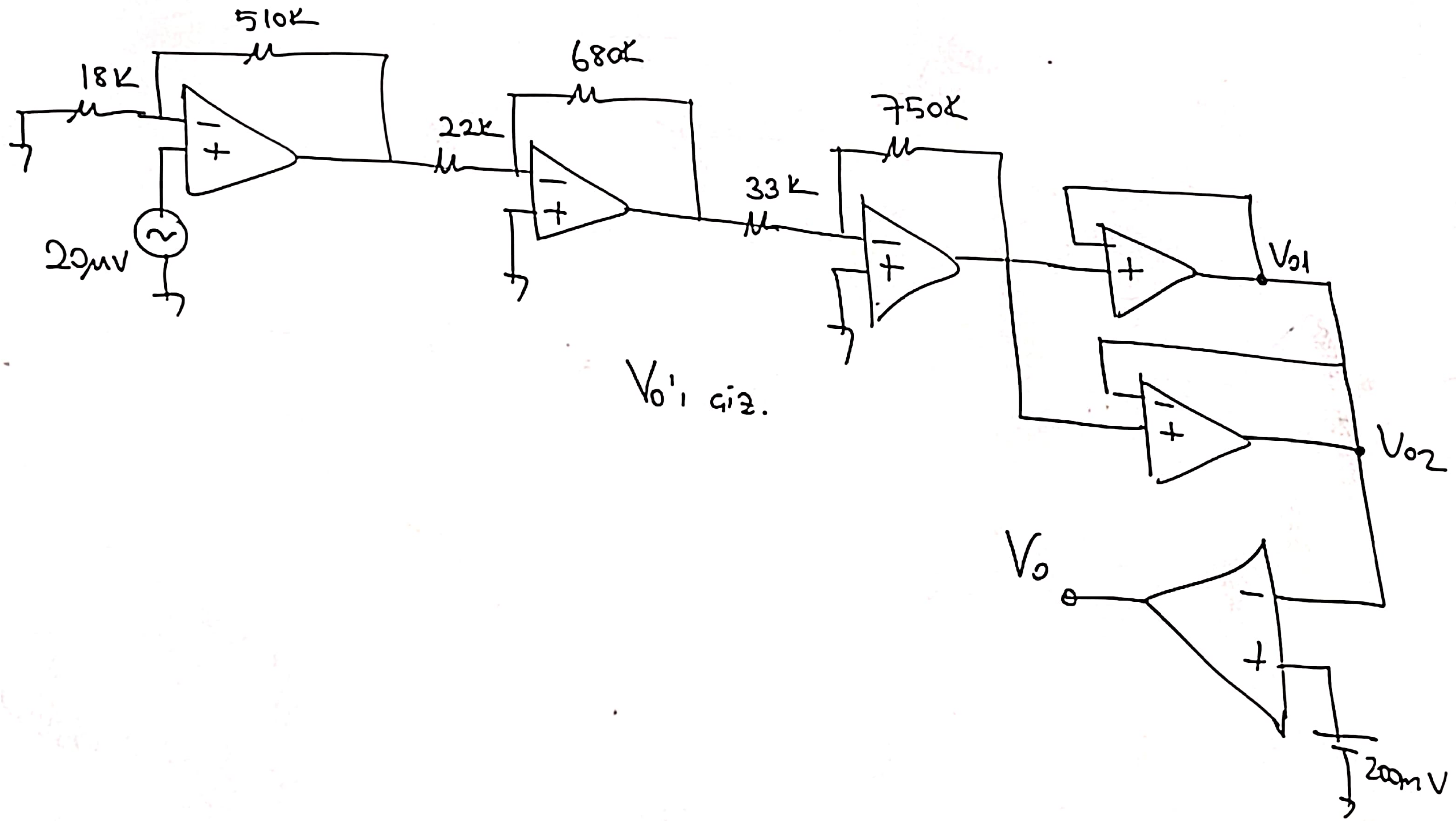
Sınav Sorusu:

$\pm V_{cc} = \pm 10V$



V_o'ı çiz.

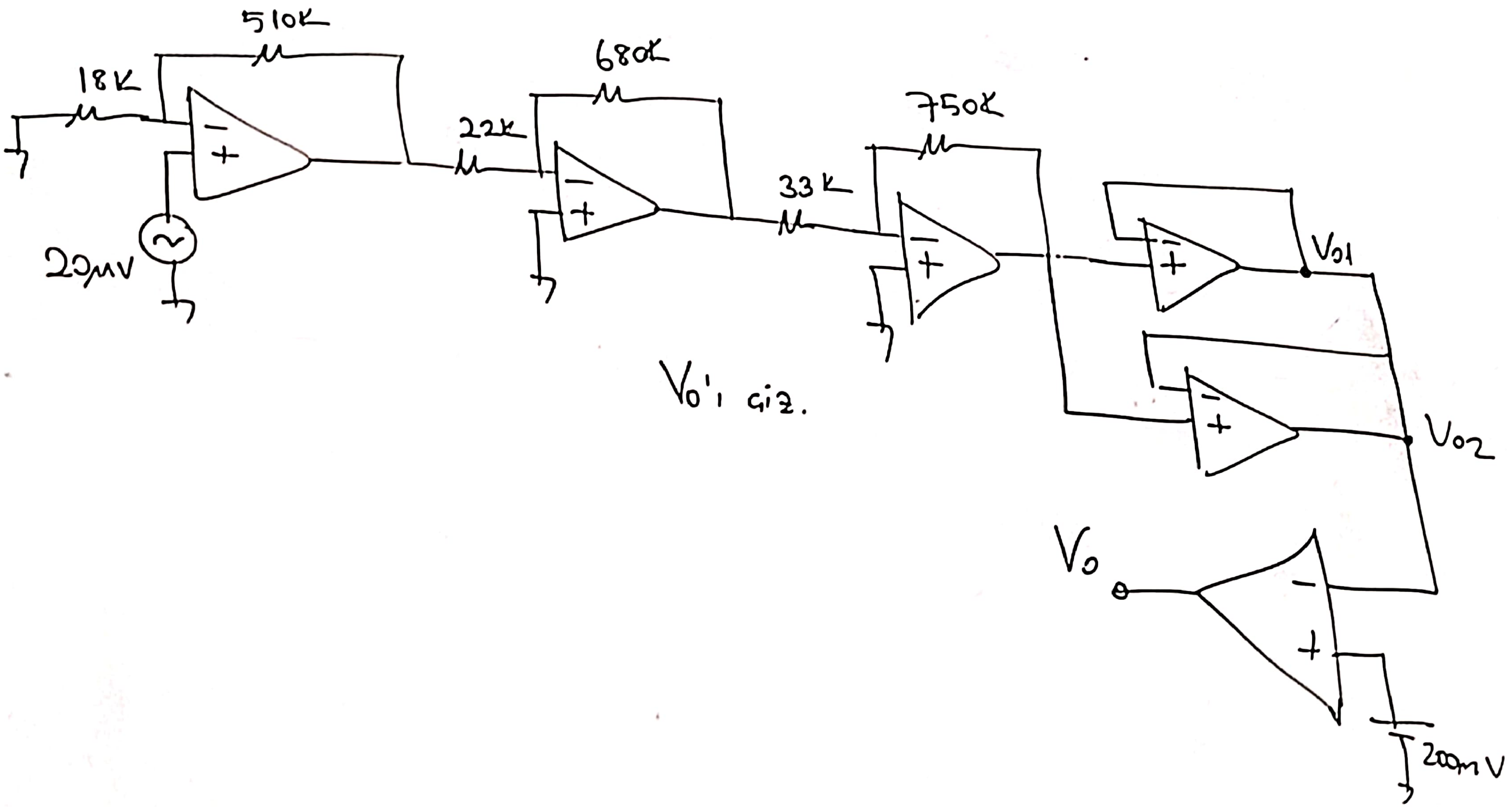
Sınav Sorusu: $\pm V_{cc} = \pm 10V$



V_o'ı çiz.

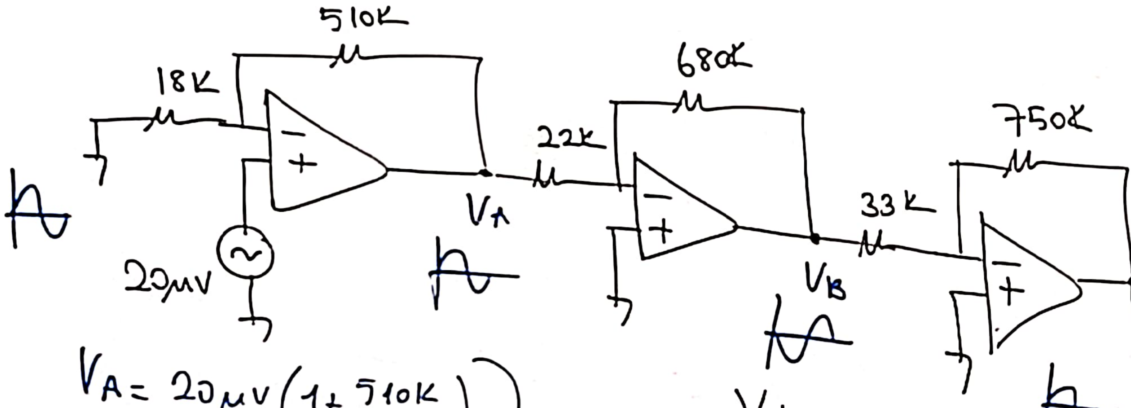
Sınav Sorusu:

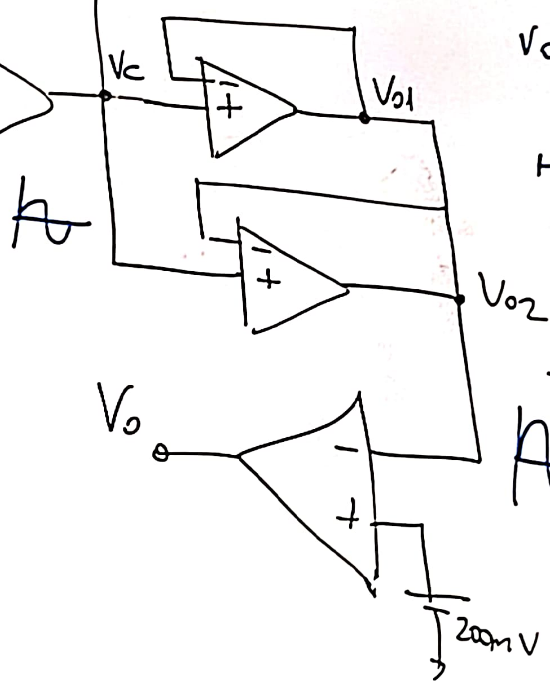
$\pm V_{cc} = \pm 10V$



V_o'ı çiz.

Sınav Sorusu: $\pm V_{CC} = \pm 10V$





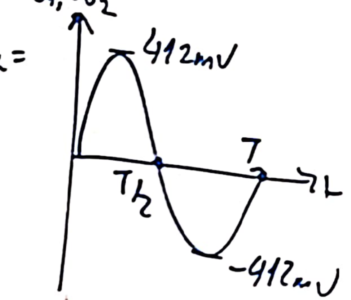
$$V_A = 20\mu V\left(1+\frac{510K}{18K}\right)$$

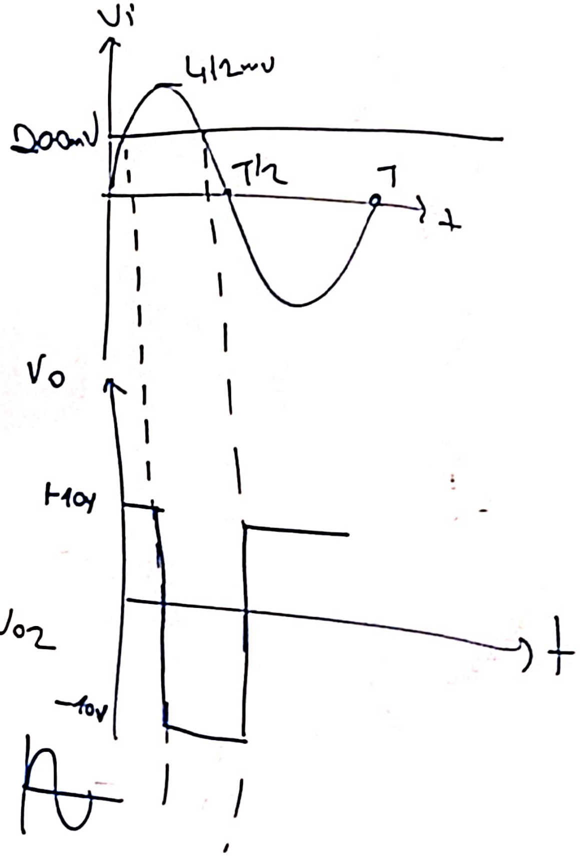
$$V_B = -V_A\cdot\left(\frac{680K}{22K}\right)$$

$$V_C = -V_B\left(\frac{750K}{33K}\right)$$

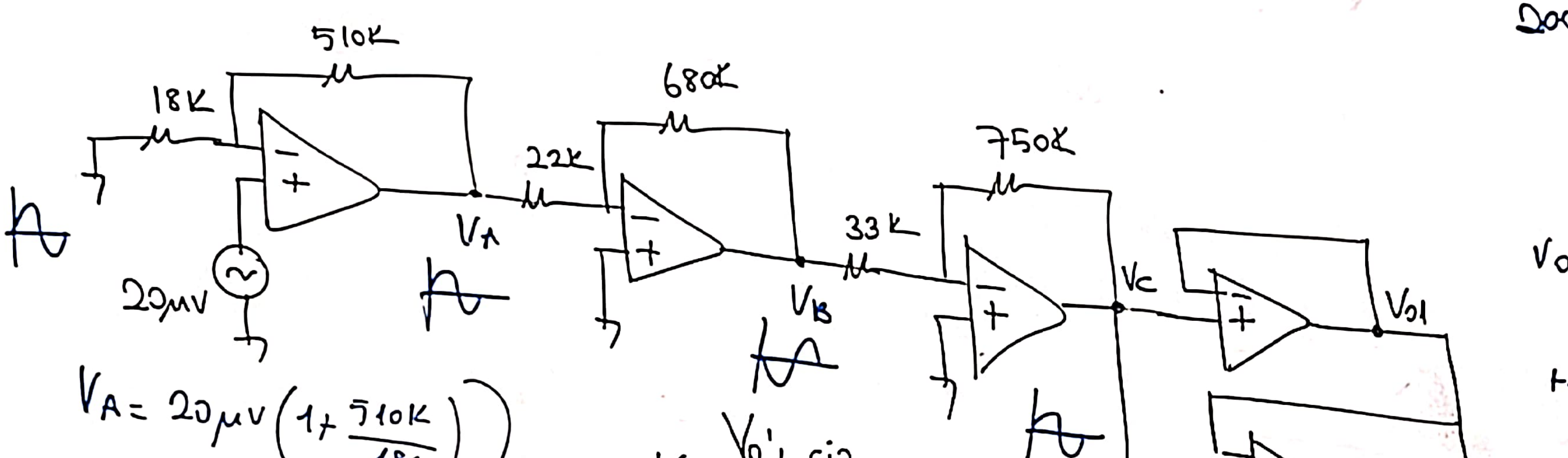
V_o'ı çiz.

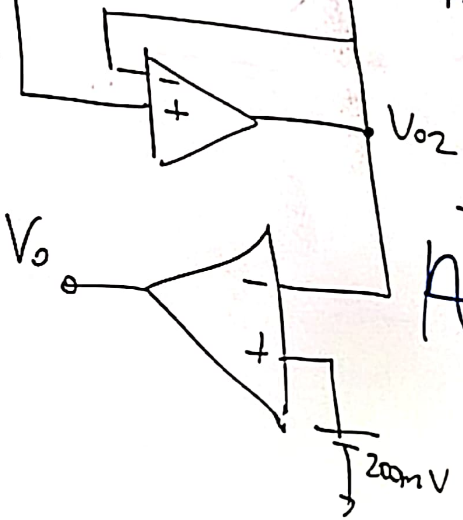
$V_{o1} = V_{o2} =$





Sınav Sorusu: $\pm V_{cc} = \pm 10V$





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V_o'ı çiz.

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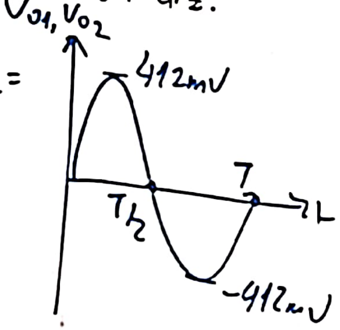
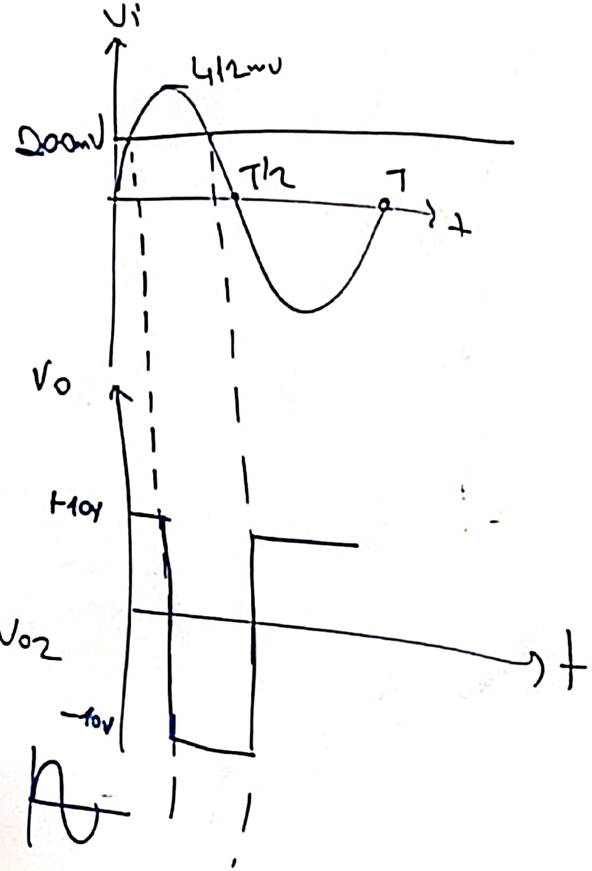
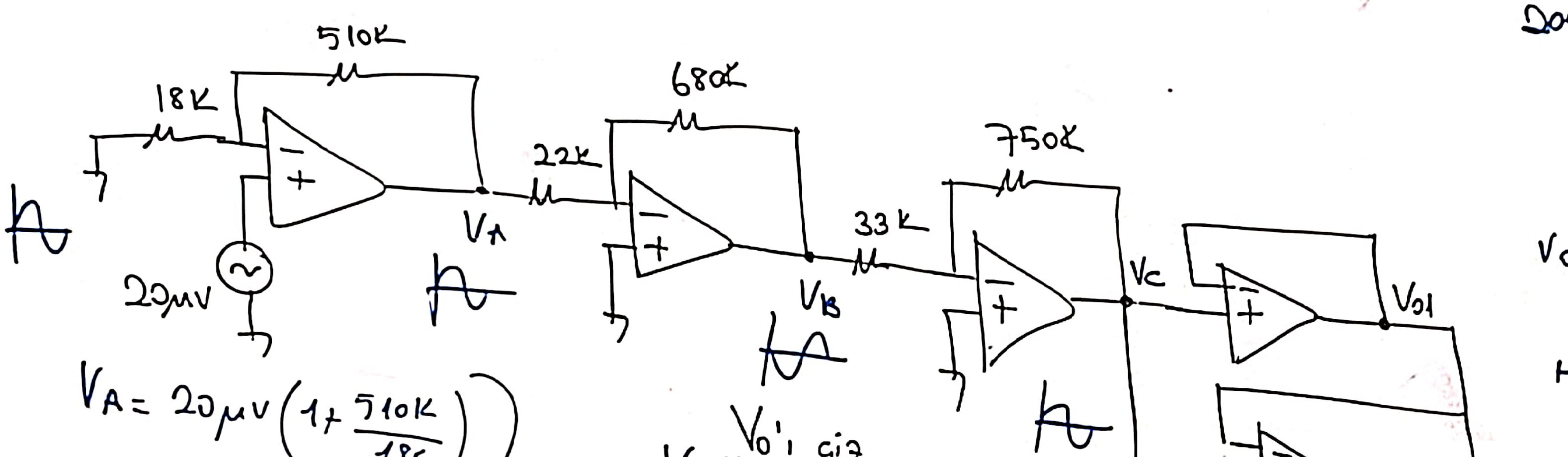
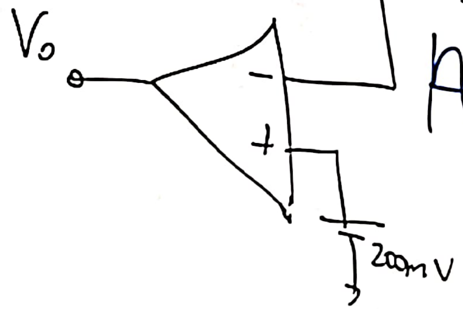




Sınav Sorusu:

$\pm V_{cc} = \pm 10V$





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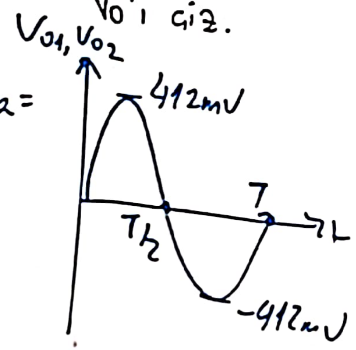
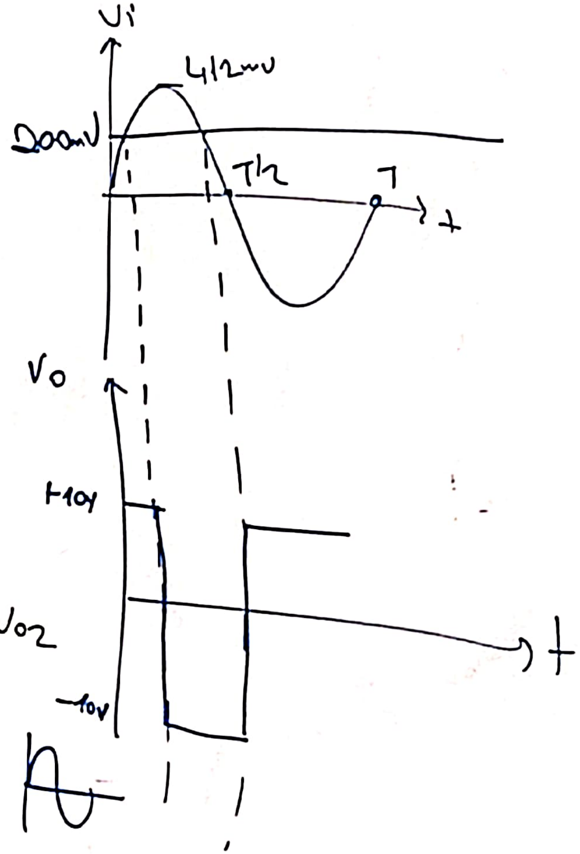
$$V_B = -V_A\left(\frac{680K}{22K}\right)$$

$$V_C = -V_B\left(\frac{750K}{33K}\right)$$

Vo'ı çiz.

$V_{01} = V_{02} =$

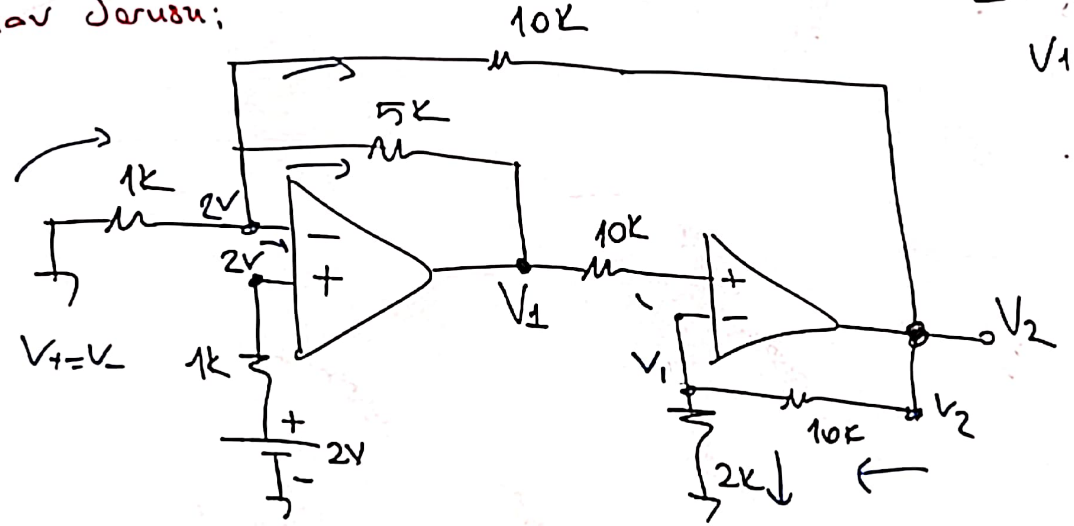




Sınav Sorusu:

$\pm V_{cc} = \pm 15V$

$V_1 = ?$ $V_2 = ?$



$$\frac{0-2}{1k} = \frac{2-V_1}{5k} + \frac{2-V_2}{10k}$$

(10) (2)

$$-20 = 4 - 2V_1 + 2 - V_2$$

$$2V_1 + \underbrace{V_2}_{6V_1} = 26$$

$$8V_1 = 26$$

$$\boxed{V_1 = 3{,}25V}$$

$$\frac{V_2 - V_1}{\cancel{10k}\ 5} = \frac{V_1}{\cancel{2k}}$$

$$V_2 - V_1 = 5V_1$$

$$V_2 = 6V_1$$

$$V_2 = 3{,}25 \times 6$$

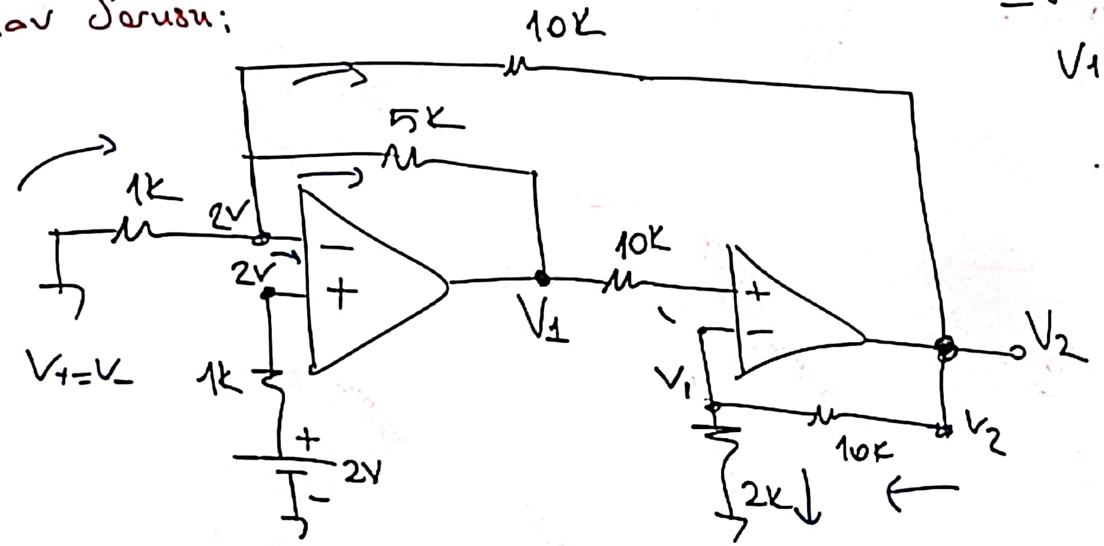
$$V_2 = 19{,}5$$

$$\boxed{V_2 = +15V}$$

Sınav Sorusu:

$\pm V_{cc} = \pm 15\,V$

$V_1 = ?\quad V_2 = ?$



$$\frac{0-2}{\underset{(10)}{1k}} = \frac{2-V_1}{\underset{(2)}{5k}} + \frac{2-V_2}{10k}$$

$$-20 = 4 - 2V_1 + 2 - V_2$$

$$2V_1 + \underbrace{V_2}_{6V_1} = 26$$

$$8V_1 = 26$$

$$\boxed{V_1 = 3,25\,V}$$

$$\frac{V_2 - V_1}{\underset{5}{\cancel{10k}}} = \frac{V_1}{\cancel{2k}}$$

$$V_2 - V_1 = 5V_1$$

$$V_2 = 6V_1$$

$$V_2 = 2,25 \times 6$$

$$V_2 = 19,5$$

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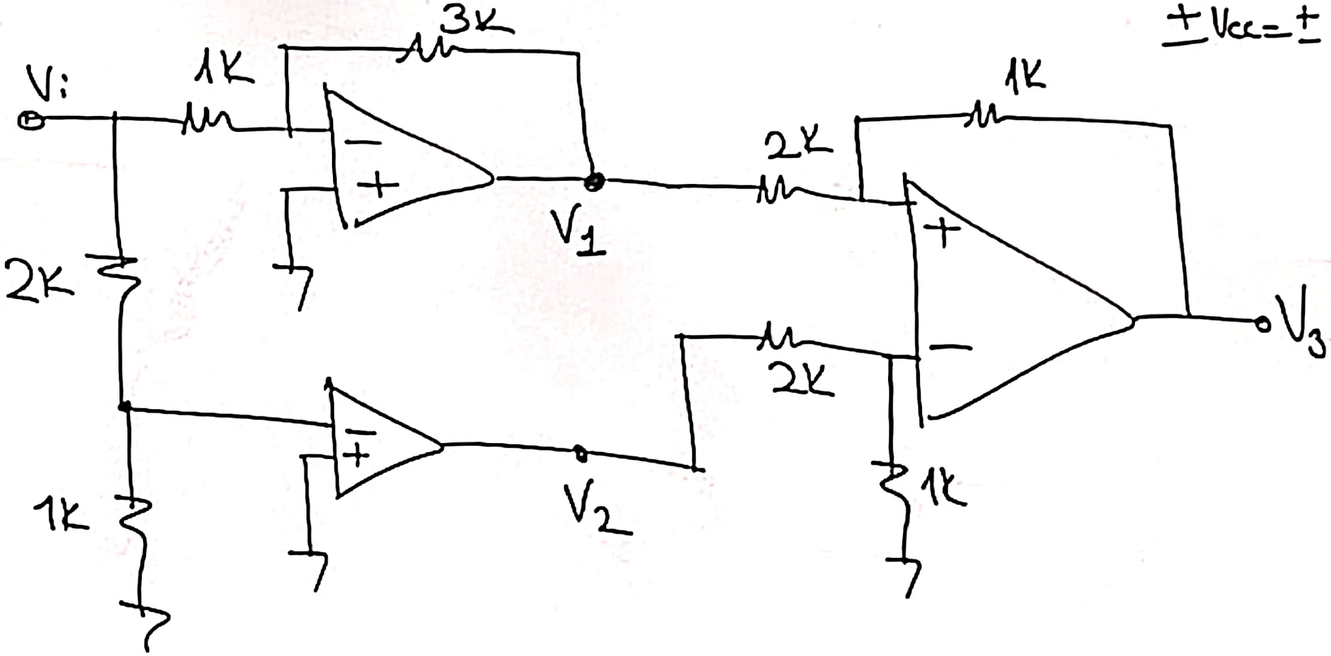
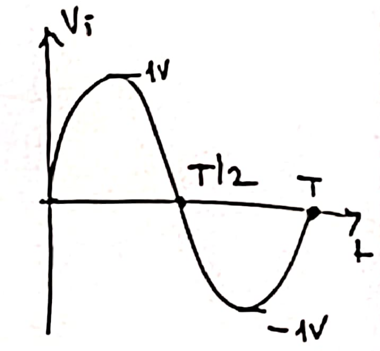
Sınav Sorusu:

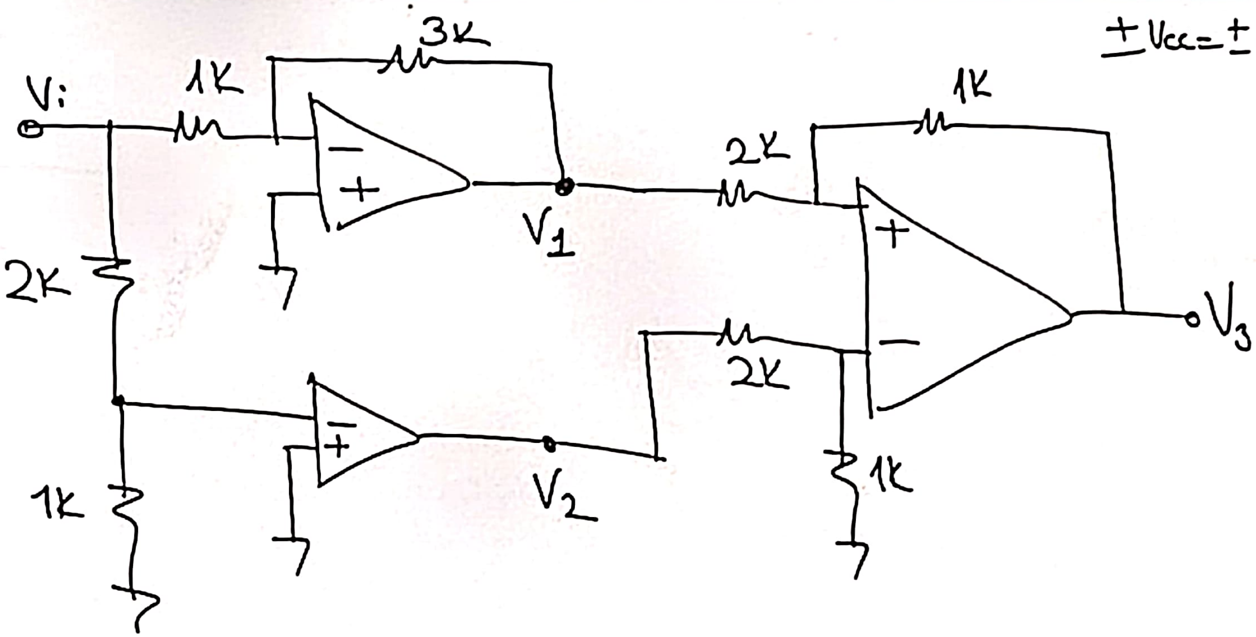


$\pm V_{cc} = \pm 12V$

V_3'ü çiziniz.

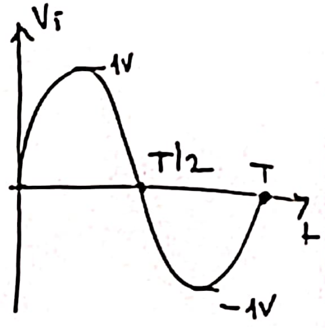


Sınav Sorusu:



$\pm V_{cc} = \pm 12V$

V_3'ü çiziniz.



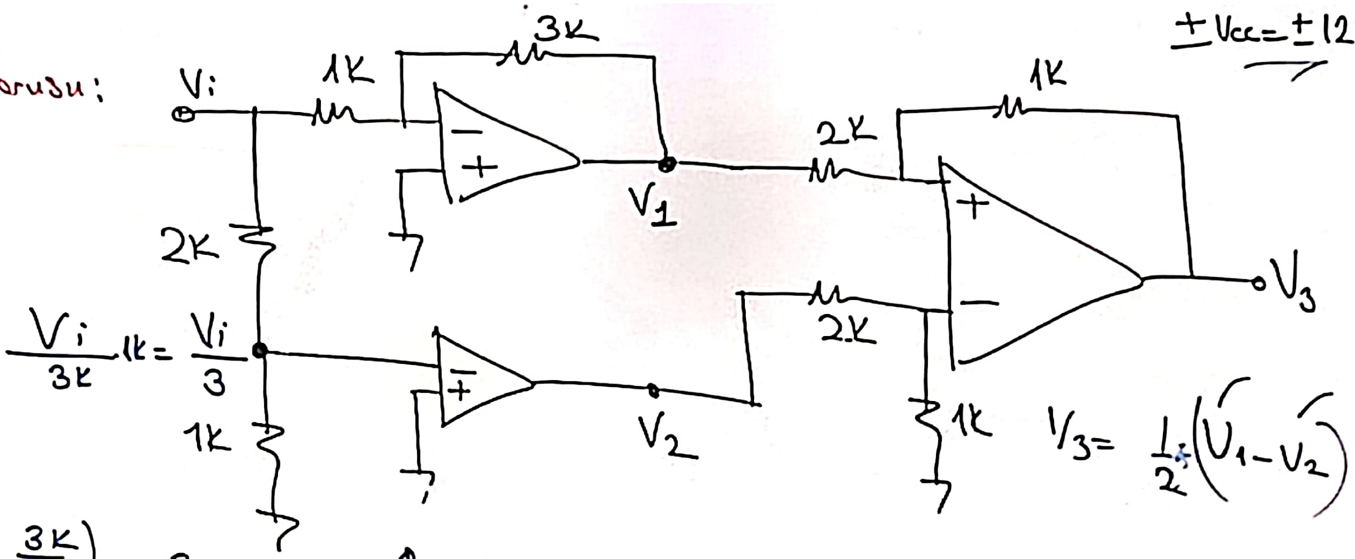
±15 V

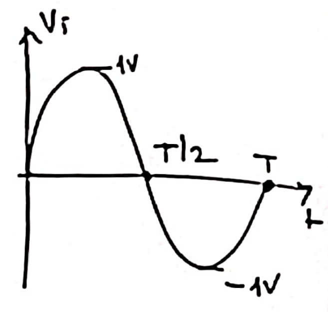
$V_2 = ?$

Sınav Sorusu:

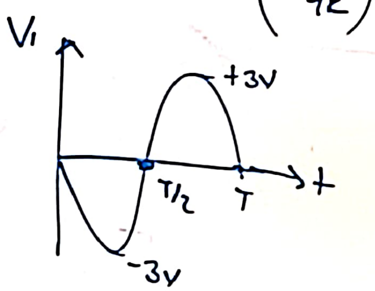
$\pm V_{cc} = \pm 12V$

V_3'ü çiziniz.

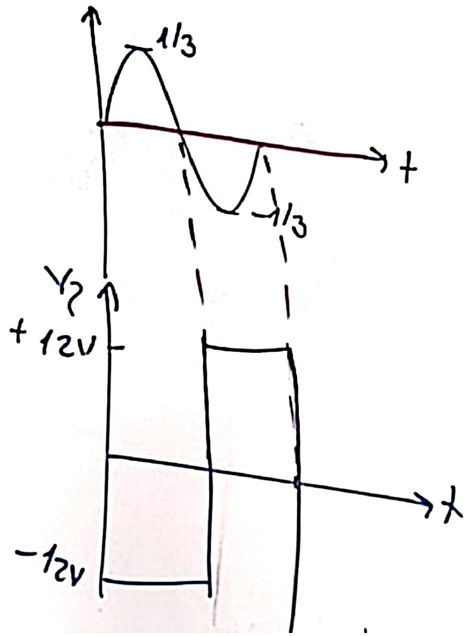




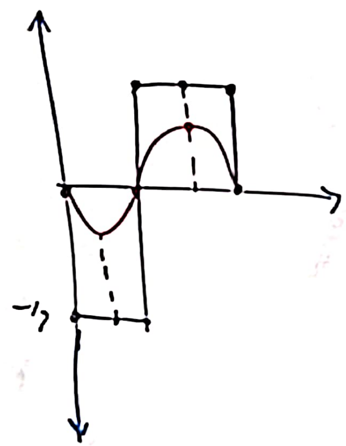
$$V_1 = -V_i \cdot \left(\frac{3K}{1K}\right) = -3V_i$$

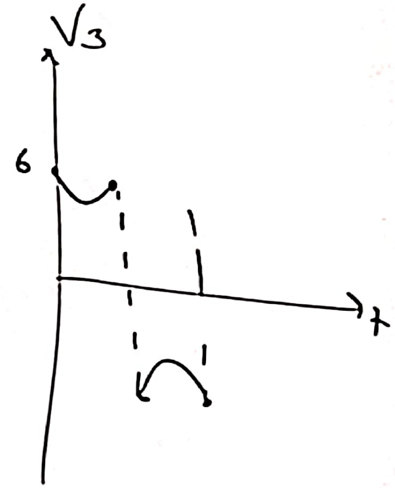


$V_2 \Rightarrow$



$$V_3 = \frac{1}{2}(V_1 - V_2)$$





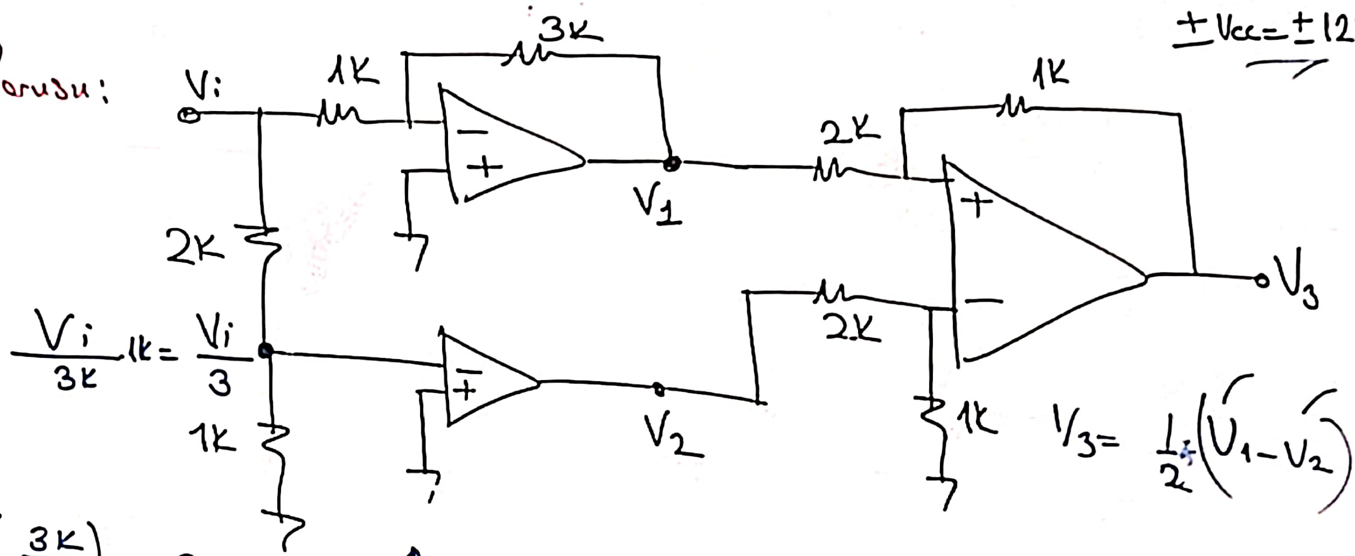
±15 V

$V_2 = ?$

Sınav Sorusu:

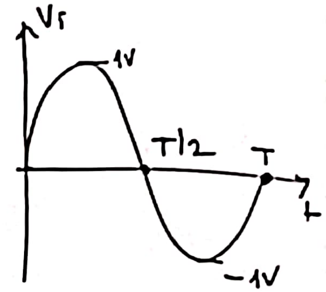
$\pm V_{cc} = \pm 12V$

V_3'ü çiziniz.



$$\frac{V_i}{3K} \cdot 1k = \frac{V_i}{3}$$

$$V_3 = \frac{1}{2}(V_1 - V_2)$$



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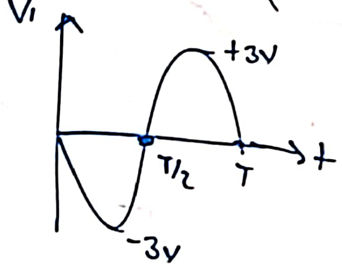
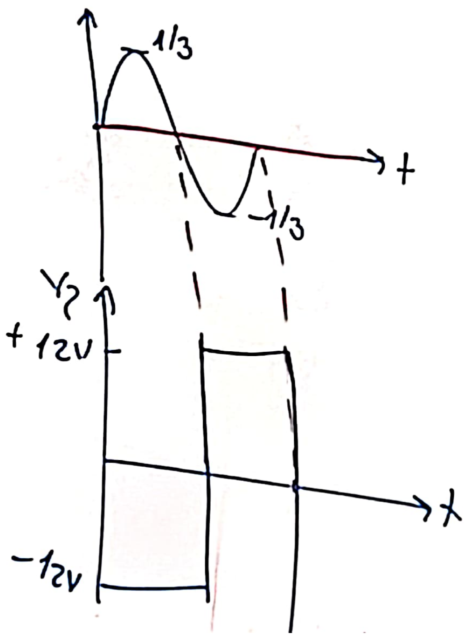


$V_2 \Rightarrow$



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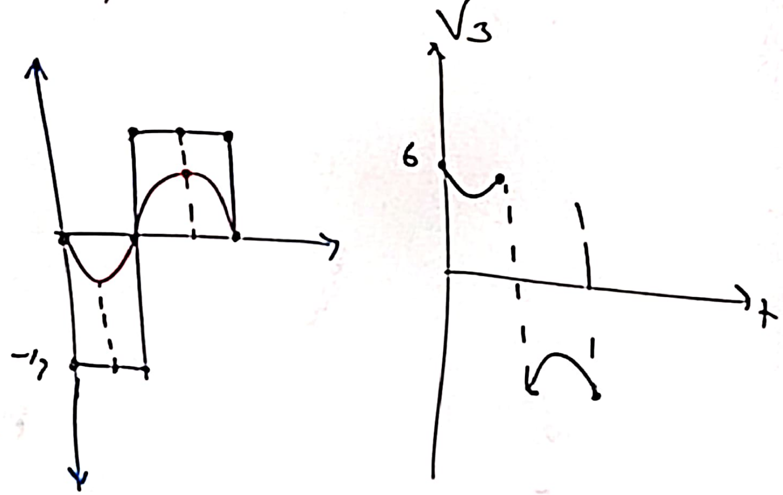
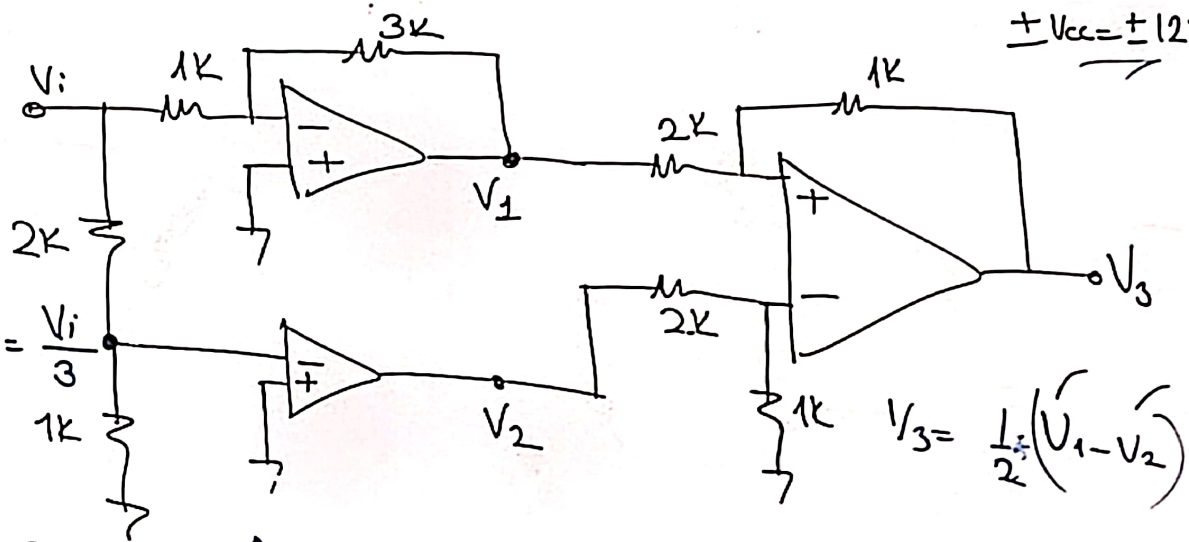


$\pm 15\,V$

? $V_2 = ?$

Sınav Sorusu:

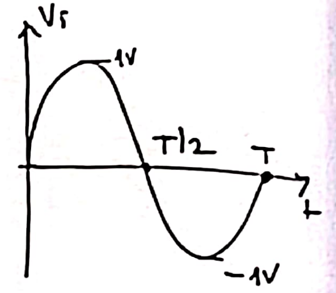


$\frac{V_i}{3K} \cdot 1K = \frac{V_i}{3}$

$V_3 = \frac{1}{2}(V_1 - V_2)$

$\pm V_{cc} = \pm 12V$

V_3'ü çiziniz.



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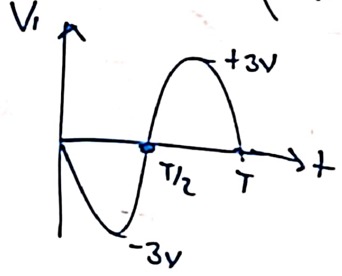

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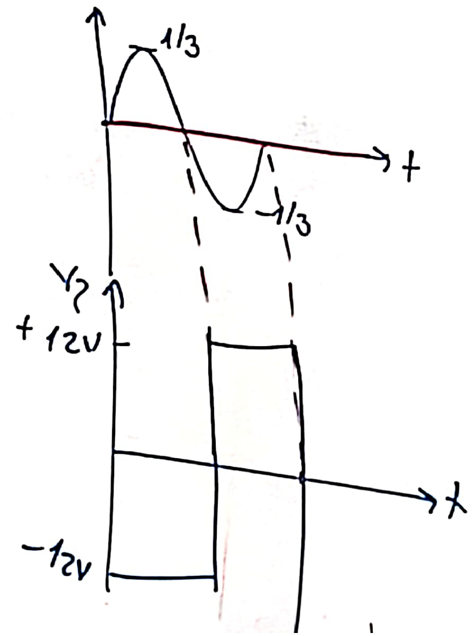

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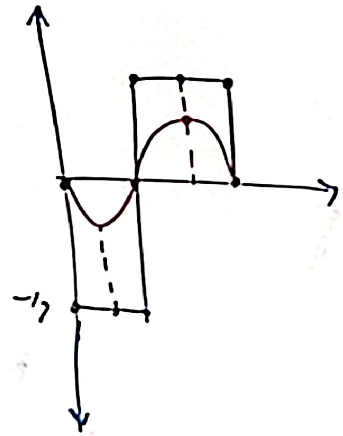

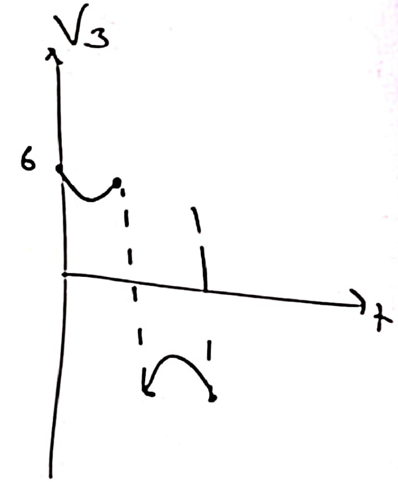

Sınav Sorusu:

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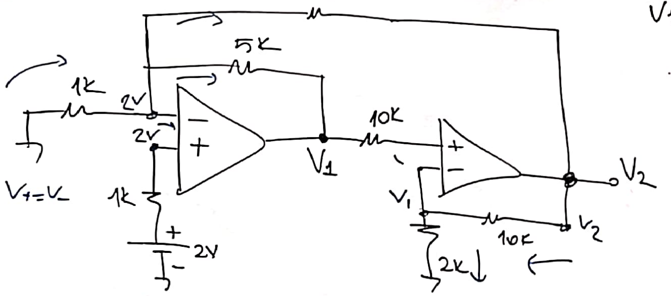
$V_1 = ?$ $V_2 = ?$



$$\frac{0-2}{1k} = \frac{2-V_1}{5k} + \frac{2-V_2}{10k}$$

(10) (2)

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Sınav Sorusu:

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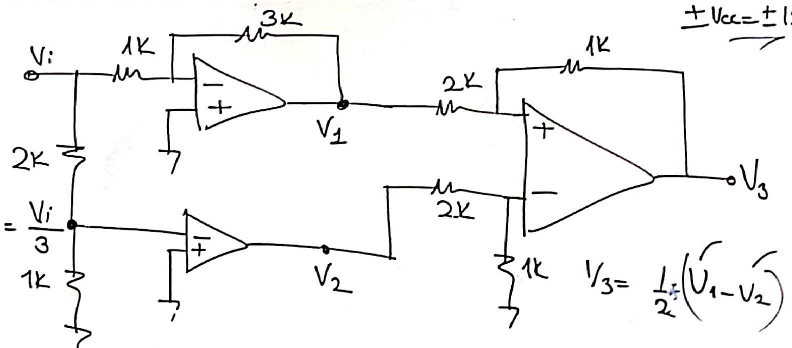
V_3'ü çiziniz.



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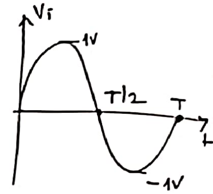

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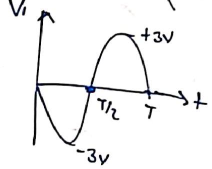

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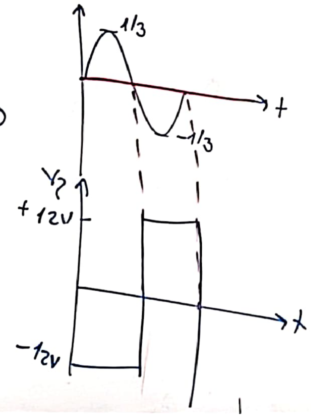

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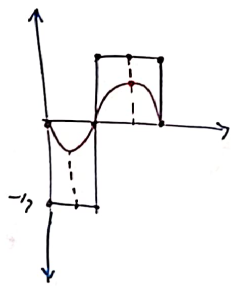
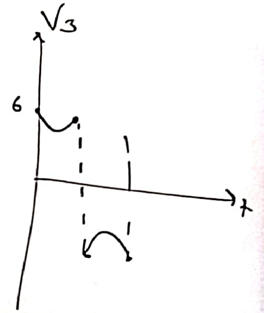
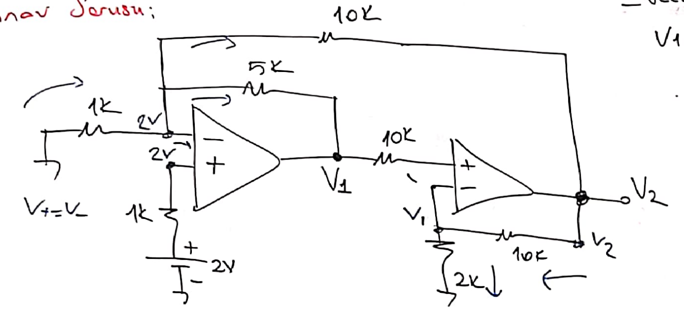




Sınav Sorusu:



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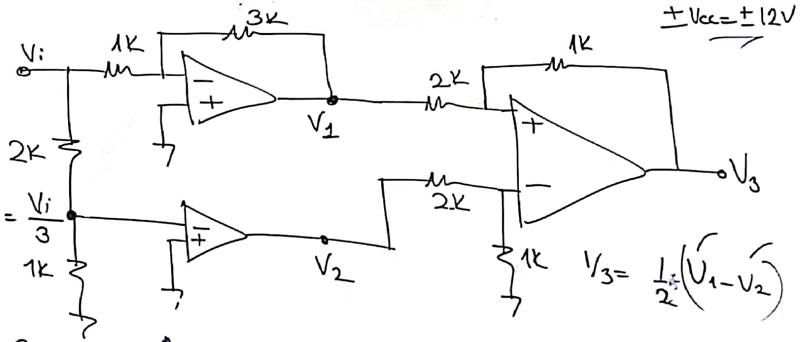

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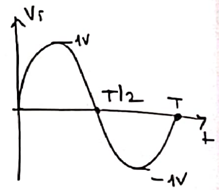
V_3'ü çiziniz.



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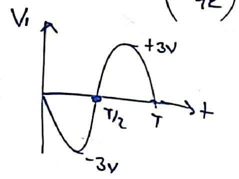
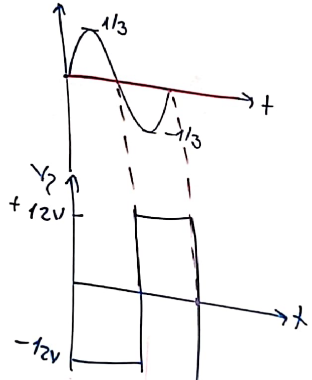


$V_2 \Rightarrow$



$$V_3 = \frac{1}{2}(V_1 - V_2)$$